

Using fish landing data to assess reef fish stocks in Koror, Palau



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Great Barrier
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PICRC Technical Report No. 25-10

April 2025

Abstract

In Palau, reef fish are a vital source of protein and have an important role in the local culture and economy. However, over time there have been increasing concerns about the sustainability of Palau's reef fisheries. In 2021, Koror state developed the Koror Southern Lagoon Coastal Fisheries Management Plan (KSLCFMP), which includes several fisheries management regulations, to assist with the recovery of reef fish populations. From 2023 to 2024, the Palau International Coral Reef Center (PICRC) conducted a 1-year survey of fish landings at the JR5 fish market in Koror to assess the status of reef fish stocks data and to provide feedback on the KSLCFMP before implementation. This was combined with previous data collected in 2014-2015 and 2016-2017. Fisher interviews conducted showed that fishers mainly went fishing in the outer and back reefs of Koror state, and the main fishing method was night spearfishing. There was also an increase in catch per unit effort (CPUE) for night spearfishing from 2017 to 2023. A substantial change in species composition was seen over time, with a higher diversity of fish being landed from 2015 to 2023 compared to 1984 and 1991. Out of the 45 species assessed using the length-based spawning potential ratio (LB-SPR) method, 14 were categorised as depleted, 25 were categorised as depleting, and six were categorised as sustainable. Most surgeon/unicornfish, groupers, emperors, snappers, and parrotfish were depleted or depleting. Mackerels and squirrelfish species were categorised as depleting. Trevallies, rabbitfish, and sweetlips were either depleting or sustainable, whereas goatfish were categorised as sustainable. Several factors can affect LB-SPR assessments including small sample sizes and inaccurate life history information. To improve these assessments, it is recommended that life history parameters are estimated for species which have uncertainties and species which are lacking this information from Palau. In addition, we recommend that the national regulations on size limits for groupers are increased, and some of the proposed minimum size limits in the KSLCFMP are amended for management to be more effective.

1. Introduction

Small-scale coral reef fisheries are a vital part of the culture, economy, and food security in many coastal communities around the world, with local populations still largely depending on subsistence fishing for their main source of protein (Sadovy, 2005). However, growing human populations, advancements in fishing technology, and the emergence of cash economies and export markets have intensified demand, increasing the pressure on reef fish populations over time (Sadovy, 2005). Furthermore, small-scale fisheries often lack the information needed for traditional stock assessments including historical data on catch and effort, as well as high quality biological information for the exploited stock (Costello et al., 2012). This means that the majority of data-poor fisheries remain un-assessed, making it difficult to effectively manage and conserve them (Costello et al., 2012).

Several methods to assess data-poor fisheries have emerged in recent years, including the length-based spawning potential ratio (LB-SPR) method (Hordyk et al., 2014), length-based indicators (LBI) (Froese, 2004), and the length-based Baeyesian biomass approach (LBB) (Froese et al., 2018). These methods only require length-frequency data, which is relatively easy and cost-effective to collect in data-poor fisheries, and a recent study found all three methods resulted in the same exploitation status for assessed stocks (Medeiros-Leal et al., 2023). The LB-SPR approach uses length composition data of a harvested fish population together with life-history parameters to estimate spawning potential ratio (SPR) and the ratio of fishing mortality rate to natural mortality rate (F/M) (Hordyk et al., 2014). SPR is a measure of the impact of fishing on the potential productivity of a stock and can be defined as the proportion of unfished reproductive potential left at any given level of fishing pressure (Goodyear, 1993). When SPR equals 1 the stock is completely unexploited and when SPR is 0 the stock has collapsed, with an SPR >0.4 indicating maximum sustainable yield and an SPR <0.2 indicating the stock is heavily exploited (Hordyk et al., 2014) (Prince et al., 2015). Many studies have used LB-SPR to assess data-poor fisheries using fishery-dependent data (e.g. Nadon et al., 2015, Filous et al., 2022, Medeiros-Leal et al., 2023), however LBSPR is heavily influenced by values of input life history parameters, particularly asymptotic length (L_{∞}) and the ratio of natural mortality to the growth coefficient (M/k) (Hordyk et al., 2015, Medeiros-Leal et al., 2023), highlighting the need for accurate locally-derived biological information.

The Republic of Palau is a small island nation located in the western Pacific Ocean. Like many Pacific Island nations, reef fisheries play a crucial role in Palau's economy, ensuring food security, and the continuation of fishing traditions (Johannes, 1981). The reef-fishery in Palau is small-scale, with the majority of landed fish used for subsistence purposes (~60%) (Lingard et al., 2011) with the rest (~40%) destined for fish markets, supermarkets, restaurants, fishing cooperatives or exported out of the country (before a reef fish export ban was implemented) (Dacks et al., 2020). This makes enforcement and monitoring of reef fish stocks highly challenging due to a large number of small boats and landing sites across the country. In addition, illegal activities, such as poaching and the exportation of fish, are known to occur. Reef fish populations are traditionally and locally managed in Palau through the use of a "Bul", which involves the implementation of a variety of controls on reef and lagoon fishing to conserve marine resources (Johannes, 1981). The first official marine conservation efforts in Palau started with the establishment of the first marine protected area (MPA) in 1956, the Ngerukewid Islands Preserve (KSG, 2012). Since then, a wide network of more than 30 MPAs have been established across the country and numerous other regulations have been implemented in response to concerns about dwindling reef resources since the 1970s (Johannes, 1981). The Marine Protection Act, passed in 1994 (amended in 2015), includes seasonal bans on groupers and rabbitfish during their spawning seasons, a complete ban on fishing for bumphead parrotfish (*Bolbometopon muricatum* - kemedukl) and Napoleon wrasse (*Cheilinus undulatus* - maml), a ban on fishing using underwater breathing apparatus, and a minimum mesh size for net fishing. The only national minimum size limits for finfish in Palau are for five species of grouper (Appendix 1). More recently, a ban on the exportation of reef fish out of the country was implemented in 2020. In 2021, the state of Koror (the most populous state in Palau) developed a new fisheries management plan, the Koror Southern Lagoon Coastal Fisheries Management Plan (KSLCFMP) (Gouezo et al., 2021), which has not yet been implemented. The plan was developed based on the Northern Reefs Fisheries Management Plan, developed in 2016, which included the implementation of temporary fishing bans as well as minimum size limits for the states of Ngarchelong and Kayangel (TNC, 2016). The KSLCFMP includes spatial and temporal measures, as well as minimum size limits for 22 reef fish species (see Appendix 1).

Life history parameters have previously been estimated for several key reef fish species in Palau (Moore et al., 2014, Prince et al., 2015, Prince, 2016b, Prince, 2016a, Prince and Lindfield, 2018, Prince et al., 2022), allowing LB-SPR to be estimated using fishery-dependent (Prince et al., 2015, Prince, 2016b, Lindfield, 2017, Lindfield, 2020) and fishery-independent data (Muller-Karanassos et al., 2021). However, only a few species have been assessed due to a lack of locally-derived life history information for many commercially important species. Other than collecting gonad and otolith samples, life history parameters can be estimated using collected fish length data or data synthesized from other studies. The Palau International Coral Reef Center (PICRC) conducted fishery-dependent monitoring across one year at the main fish market in Koror, Palau from February 2023 to February 2024. This data was combined with previous fish landing data collected in 2014-2015 (Moore et al., 2014, Lindfield, 2017), and 2016-2017 (unpublished data). Life history parameters were collated and estimated to 1) assess how species composition of reef fish landings have changed over time, 2) assess the status of 45 commercially important reef fish species using the LBSPR method, and 3) propose management recommendations and identify future research needs for the reef fishery.

2. Methods

2.1 Species length composition and fisher interviews

PICRC conducted fish landing surveys at the JR5 Fish Market in Koror, Palau, across the span of one year from February 2023 to February 2024. A stereo video system, consisting of two GoPro Hero 4 cameras in waterproof housings mounted on a table was used to record fish landings (Figure 1). Daily surveys were conducted in the morning from around 6am to 8am every day from Monday to Friday, as well as one to two Saturdays per month. Fish market surveys were also conducted by PICRC, in collaboration with the University of Guam and The Nature Conservancy, from November 2016 to November 2017. In 2016-2017, a Fujifilm 3D-W3 stereo-camera was used to record fish landings. Daily surveys were conducted in the morning from around 6am to 8am and in the evening from around 3:30pm to 5:30pm on alternate days from Monday to Friday, as well as on Saturday or Sunday every other week. Fisher interviews were also conducted in 2017 and 2023, to obtain information on fishing state, fishing methods, habitat, number of fishers, time spent fishing, and total weight of the catch.

The Bureau of Fisheries (BOF), under the Ministry of Agriculture, Fisheries, and the Environment (MAFE) in Palau collected reef fish length data in 2014 and 2015. Most data were collected at the JR5 fish market and includes data collected in September 2014 as part of a training program run by the Secretariat of the Pacific Community (SPC) (Moore et al., 2014) as well as additional data collected over a 3-month period from 11th May to 14th August 2015 using a Fujifilm 3D-W3 stereo-camera (Lindfield, 2017). Additional data was collected at the Palau Sports Fishing Association (PSFA) annual fishing tournament, the Etpison Cup, in 2014 and 2015.

All stereo videos were analysed using the SeaGIS EventMeasure software. Left and right videos were imported into EventMeasure and synchronized based on hand signals made at the beginning of each video. All landed fish were identified to the lowest possible taxonomic level and fork length (FL) was measured, unless the fish was distorted, partially covered, or cut out of the video. PICRC's stereo video system was calibrated before the start of the survey in February 2023. Length data were compiled from 2014 to 2024 and species which had 100 length measurements or more, were selected for analysis, resulting in a total of 45 species that were able to be assessed.



Figure 1. Stereo video system mounted on a folding table for data collection at the JR5 fish market in 2023-2024 (A) and screenshot of chum (*Naso unicornis*) from one of the videos taken at the market (B).

2.2 Historic data

Historical data is available from the 1980s and 1990s collected by the Division of Marine Resources at the Palau Federation of Fishing Association (PFFA) fish market located in Malakal, Koror, with some additional data collected at several other smaller markets (Perron et al., 1983, Kitalong and Dalzell, 1994). In the 1980s and 1990s, fork length was measured to the nearest 0.1 cm and lengths were then aggregated to 1 cm length classes. Since the 2014-2015, 2016-2017 and 2023-2024 data were recorded to the nearest millimetre, historical lengths were increased by 0.5 cm so they would be the midpoint of the 1 cm length class. Data on herbivorous fish was also collected at the JR5 market by Dr. Sonia Bejarano from the University of Queensland in March, April, and July 2009 (Bejarano et al., 2013a, Bejarano et al., 2013b). Total length data was converted into fork length using conversion values available from FishBase, apart from *Siganus doliatus*, which was converted based on total length measurements taken in 2023/2024 (see Appendix 2). Data from 2009 for *Naso lituratus* (cherangel) and *Naso unicornis* (chum) was not included since total length included caudal filaments and could not be converted to fork length. Historical data from the 1980s, 1990s, and 2009 were not included in LB-SPR assessments but were used to estimate life history parameters and assess changes in species composition.

2.3 Synthesis of life history information

Due to the limited availability of locally-derived life history information, life history parameters for each of the 45 species were obtained from various sources (Appendix 3). This included locally derived information available from Palau (Moore et al., 2014, Prince et al., 2015, Prince, 2016a, Prince, 2016b, Prince and Lindfield, 2018, Prince et al., 2022), Powell Wetherall plots, empirical equations (Froese and Binohlan, 2000, Prince et al., 2014), synthesized data at the species and family level (Prince et al., 2015, Prince, 2016b, Prince et al., 2023), FishBase (Froese and Pauly, 2024), and the Beverton-Holt invariant $M/k=1.5$ (Jensen, 1996) (Table 1).

L_{inf} (asymptotic length) was estimated for each species using Powell Wetherall plots with the TropFishR package in R (Mildenberger and Taylor, 2016) and fish length data collected in Palau from 1982 to 2024 (Perron et al., 1983, Kitalong and Dalzell, 1994, Bejarano et al., 2013a, Bejarano et al., 2013b, Moore et al., 2014, Lindfield, 2017). L_{∞} , length at 50% maturity (L_{50}), and length at 95% maturity (L_{95}) were estimated using

empirical equations (Froese and Binohlan, 2000, Prince et al., 2014) based on the 99th percentile length (L_{99th}) obtained during fish landing surveys from 1982 to 2024. The 99th percentile was used instead of the maximum length due to some uncertainty in historic data including the presence of large fish which may have been misidentified. L_{inf} , L_{50} , natural mortality (M), and von Bertalanffy growth coefficient (k) were extracted from FishBase using the rfishbase package in R (Boettiger et al., 2012), and L_{95} was estimated using the empirical equation $L_{95}=1.1*L_{50}$ (Prince et al., 2014). Values of L_{inf} that were less than the L_{∞} estimated using the Froese and Binohlan (2000) equation were deemed to be too low and removed from further analysis.

Table 1. Sources of life history values used in the LBSPR analysis

	Locally derived	Our data	Empirical equation	Synthesized data	FishBase	Beverton-Holt invariant
L_{inf}	Moore et al. (2014)	Powell-Wetherall plots	$\log(L_{inf}) = 0.044 + 0.9841 \times \log(L_{99th})$ (Froese and Binohlan, 2000)	Prince et al. (2015), Prince (2016b)	FishBase	-
M/k	Moore et al. (2014) – estimated from M and k values	-	-	Species level (Prince et al., 2015, Prince, 2016b) and family level (Prince et al., 2023)	FishBase – estimated from M and k values	1.5 (Jensen, 1996)
L_{50}	(Prince et al., 2015, Prince, 2016a, Prince, 2016b, Prince and Lindfield, 2018, Prince et al., 2022)	-	$\log(L_{50}) = 0.8979 \times \log(L_{inf}) - 0.0782$ (Froese and Binohlan, 2000)	-	FishBase	-
L_{95}	(Prince et al., 2015, Prince, 2016a, Prince, 2016b, Prince and Lindfield, 2018, Prince et al., 2022)	-	$L_{95} = 1.1 * L_{50}$ (Prince et al., 2014)	-	FishBase $L_{95} = 1.1 * L_{50}$ (Prince et al., 2014)	-

2.4 Data analysis

For the fisher interview data from 2017 and 2023, the percentage of fishing trips per state, fishing method, and habitat were calculated and plotted using stacked bar charts. Catch per unit effort (CPUE) was calculated for each night spearfishing trip in 2017 and 2023 using the formula:

$$CPUE = \frac{Catch}{Effort}$$

Where Catch = total weight of catch (lbs) and Effort = hours spent fishing*number of fishers. Mean CPUE was compared between years using an ANOVA with log-transformed data to conform to normality assumptions. A linear model was used to assess the relationship between catch and effort.

To assess how the composition of fish landings has changed over time, bar plots were made showing species contribution to landings by abundance and biomass from 1984 to 2023. To assess the size distribution of species length-frequency histograms were plotted for the 45 species which had >100 length measurements from 2014 to 2024.

A total of 75 LB-SPR scenarios were generated using a combination of available life history parameters (Appendix 4). Scenarios were run for each of the 45 species using the LBSPR package in R (Hordyk, 2021). The best scenario was then selected for each species based on the best life history information available. Life history parameters collected from Palau were prioritised (Moore et al., 2014, Prince et al., 2015, Prince, 2016b, Prince, 2016a, Prince and Lindfield, 2018, Prince et al., 2022), followed by empirical data (Powell and Froese), synthesized data (Prince et al., 2015, Prince et al., 2023), and lastly values from FishBase. Stocks were then classified into four categories: 1) Sustainable (SPR>0.2 and FM<1), 2) Depleting (SPR>0.2 and FM>1), 3) Recovering (SPR<0.2 and FM<1), and 4) Depleted (SPR<0.2 and FM>1).

Immature fish haven't had a chance to reproduce yet, and if they are caught before they spawn the population can't replenish itself (Vasilakopoulos et al., 2011). Minimum size limits are usually set above L_{50} to protect immature individuals, with 1.1 or 1.2 x L_{50} able to sustain stocks (Prince and Hordyk, 2018). Updated minimum size limits were calculated by multiplying the most recent L_{50} values by 1.1 and 1.2 and compared to current and proposed size limits from the KSLCFMP (Appendix 1).

3. Results

3.1 Fisher interviews

Most fishing trips took place in Koror state in both 2017 and 2023 (>70% in each year), with the rest in other states (Figure 2A). In both years, night spearfishing was the main fishing method used by fishers with 59.6% of trips using night spearfishing as the

primary fishing method in 2017 and 81.1% in 2023 (Figure 2B). More fishers used a handline in 2017 (17.5%) compared to 2023 (0.9%), however this is likely due to the timing of the survey being conducted in both the morning and afternoon in 2017 compared to only in the morning in 2023, since line fishing is often conducted during the day with catches sold to the market in the afternoon. Most fishers went fishing in the back reef (46.8% in 2017 and 24.6% in 2023) and outer reef (30.4% in 2017 and 30.3% in 2023) habitats (Figure 2C). CPUE for night spearfishing increased significantly from 10.9 ± 1.1 lbs/hr in 2017 to 15.2 ± 1.1 lbs/hr in 2023 (ANOVA: $p=0.009$) (Figure 3A). There was also a significant relationship between catch and effort for night spearfishing in both 2017 and 2023 (LM: $p<0.001$), with no difference found between years (Figure 3B).

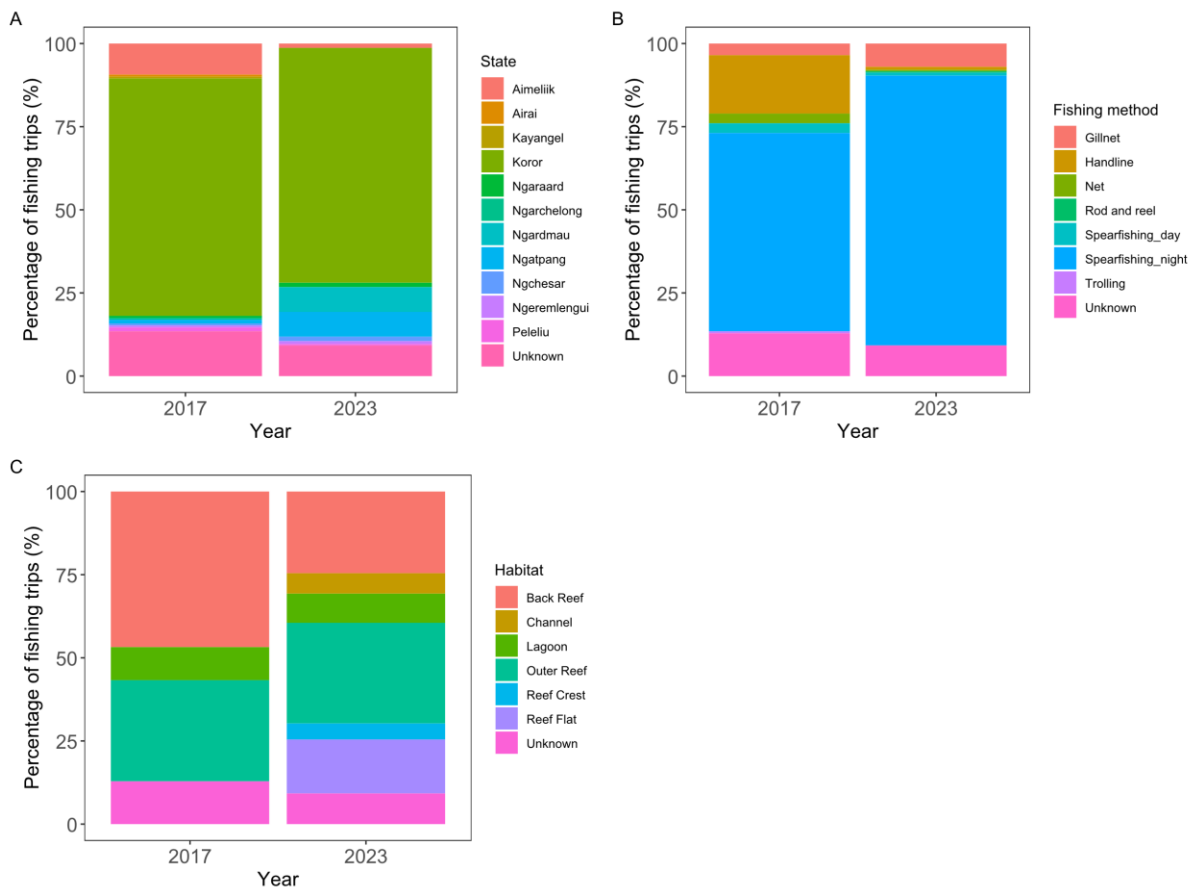


Figure 2. Stacked bar charts showing percentage of fishing trips per state (A), fishing method (B), and habitat (C) in 2017 and 2023.

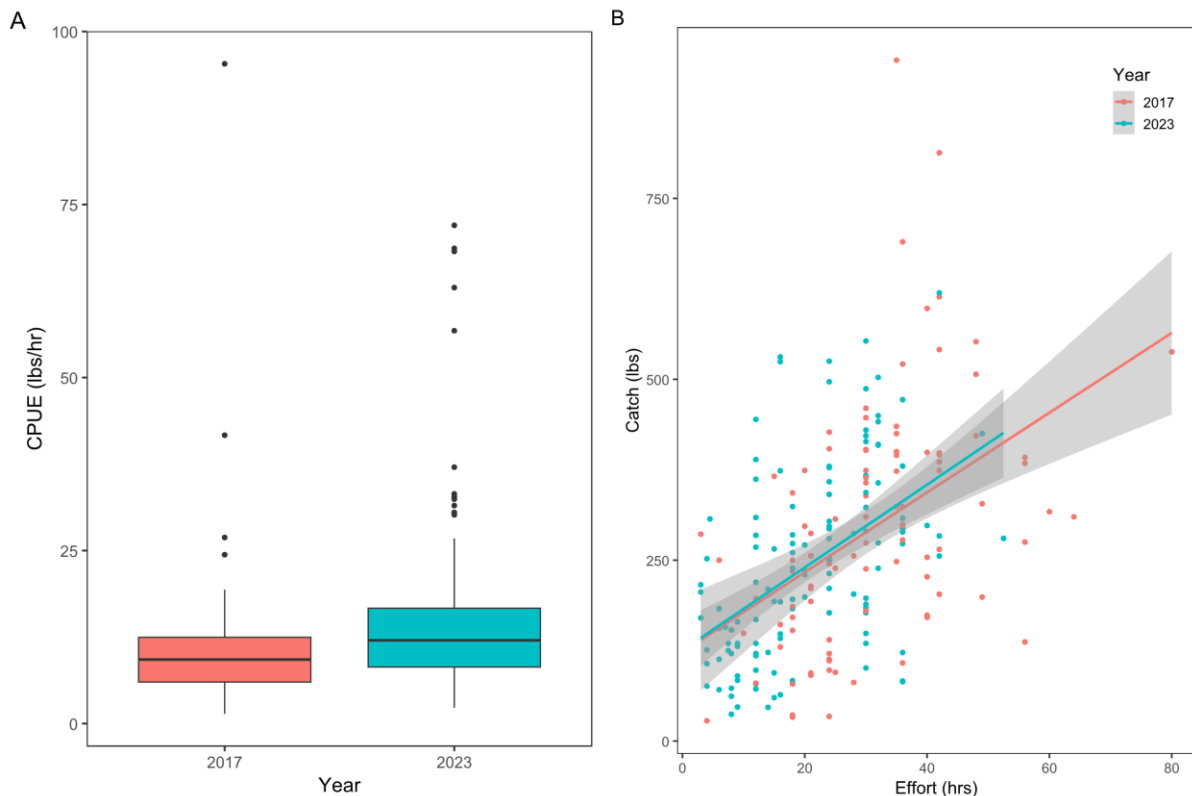


Figure 3. Catch per unit effort (CPUE) for night spearfishing in 2017 and 2023 (A) and relationship between catch and effort for night spearfishing in 2017 and 2023 (B).

3.2 Species composition

The total number of species recorded during fish landing surveys increased over time, with only 15 species recorded in 1984, 39 in 1991, 140 in 2015, 172 in 2017, and 138 in 2023. In 1984, *Epinephelus polyphekadion* (ksau temekai) made up the highest percentage of fish landings by abundance, followed by *N. unicornis*, *Scarus oviceps* and *Hipposcarus longiceps* (ngiaoch, berkism) (Figure 4A). *B. muricatum* made up the highest percentage of fish landings by biomass, followed by *N. unicornis* and *E. polyphekadion* (Figure 4B). In 1991, *Siganus fuscescens* (meyas), which has been used interchangeably with *Siganus canaliculatus* (meyas) (Hsu et al., 2011), dominated fish landings, contributing more than 15% to total fish landings by abundance. However, *B. muricatum* still had the highest contribution to total biomass, in addition to *N. unicornis* and *C. undulatus*. From 2015 to 2023, *H. longiceps* made up the highest percentage of fish landings by abundance, followed by *Lutjanus gibbus* (keremlal), *N. lituratus*, *N. unicornis*, and *Parupeneus barberinus* (bang). Total biomass was dominated by *N. unicornis* and *H. longiceps* in all three years.

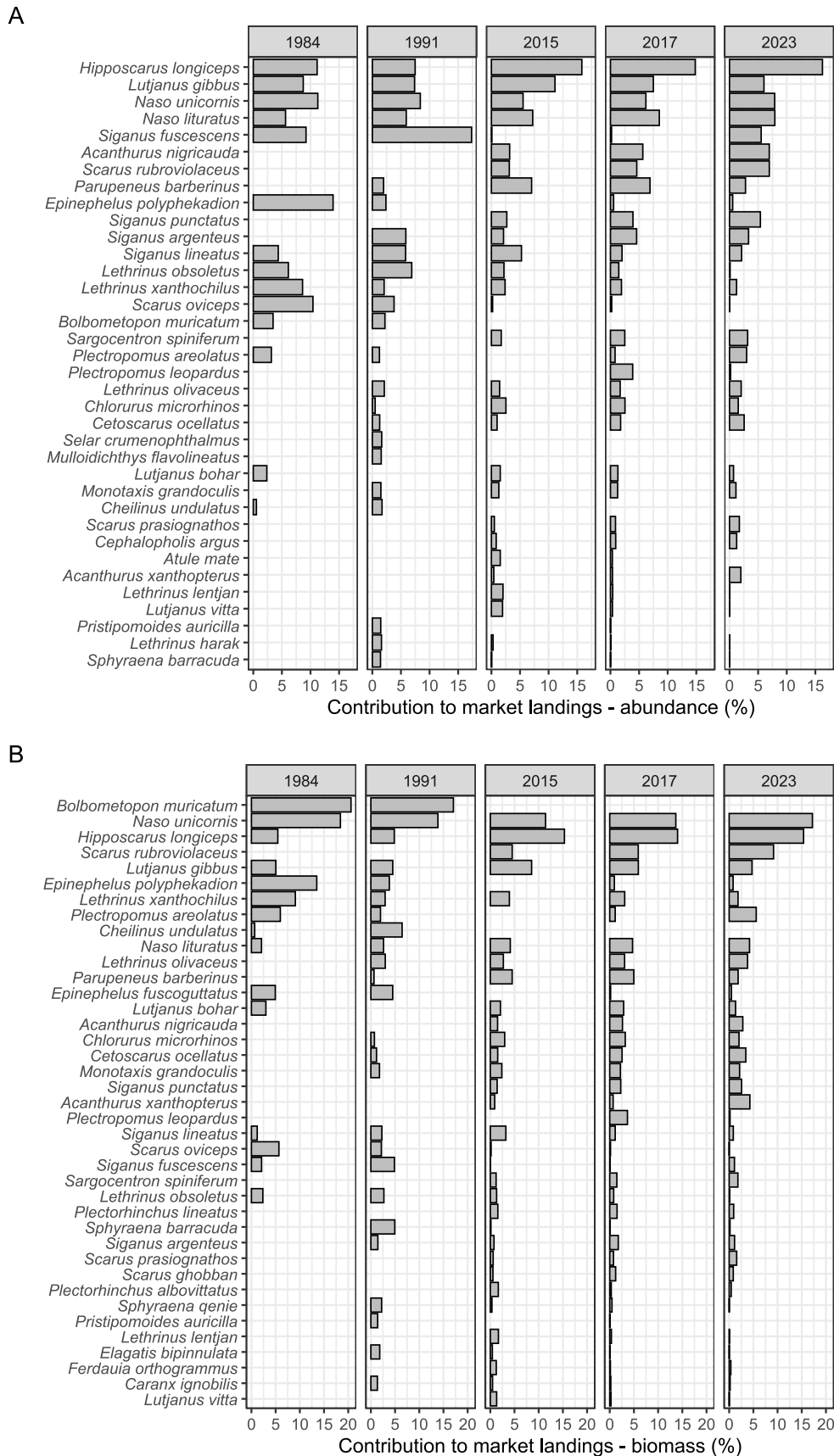


Figure 4. Species contribution (>1%) to market landings by abundance (A) and biomass (B) from 1984 to 2024. Data from 2009 is not shown since only herbivores were counted.

3.3 Length based spawning potential ratio (LB-SPR) assessments

Out of the 45 species assessed using the LB-SPR method, 14 were categorised as depleted, 25 were categorised as depleting, and six were categorised as sustainable (Figure 5). All species of Acanthuridae (surgeonfish/unicornfish), Serranidae (groupers), Lethrinidae (emperors), Lutjanidae (snappers), and Scaridae (parrotfish) were categorised as either depleted or depleting. Scombridae (mackerels) and Holocentridae (squirrelfish) species were categorised as depleting. Carangidae (trevallies), Siganidae (rabbitfish), and Haemulidae (sweetlips) were either depleting or sustainable. Whereas Mullidae (goatfish) were categorised as sustainable. Only one species (*Parupeneus cyclostomus* - bang) had an SPR >0.4 and an F/M <0.5.

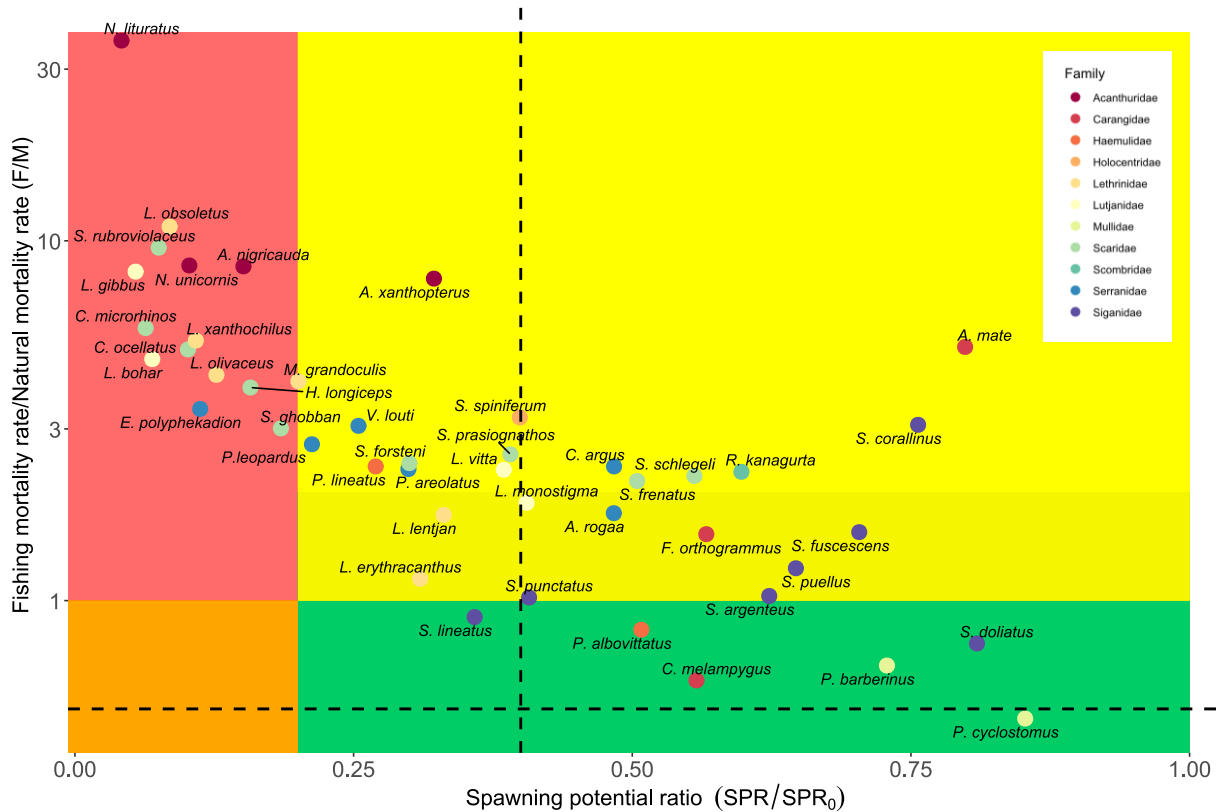


Figure 5. Scatter plot showing best estimates of spawning potential ratio (SPR/SPR₀) and fishing mortality rate/natural mortality rate (F/M) estimated from the LB-SPR assessment for 45 commercially important reef fish species in Palau from 2014 to 2024. The red area shows stocks that are being overfished, the yellow area shows stocks that are depleting, the orange area shows stocks that are recovering, and the green area shows stocks that are considered sustainable. Black dashed lines indicate target reference points of 0.5 F/M and 0.40 SPR.

Acanthuridae (surgeonfish/unicornfish)

The percentage of immature fish landed from 2014 to 2024 was 4.8% for *Acanthurus nigricauda* (chesangel), 0.3% for *N. lituratus*, and 24% for *N. unicornis* (Figure 6). There is no locally derived size of maturity estimate for *Acanthurus xanthopterus* (mesekuuk). Three out of the four assessed Acanthuridae species had best SPR estimates below 0.2, including *A. nigricauda* – 0.15 (0.14-0.16 95% CI), *N. lituratus* – 0.04 (0.04-0.04 95% CI) and *N. unicornis* – 0.10 (0.08-0.12 95% CI). *A. xanthopterus* had an SPR of 0.32 (0.25 – 0.39 95% CI) (Figure 7, Appendix 5). All Acanthuridae species had an F/M above one.

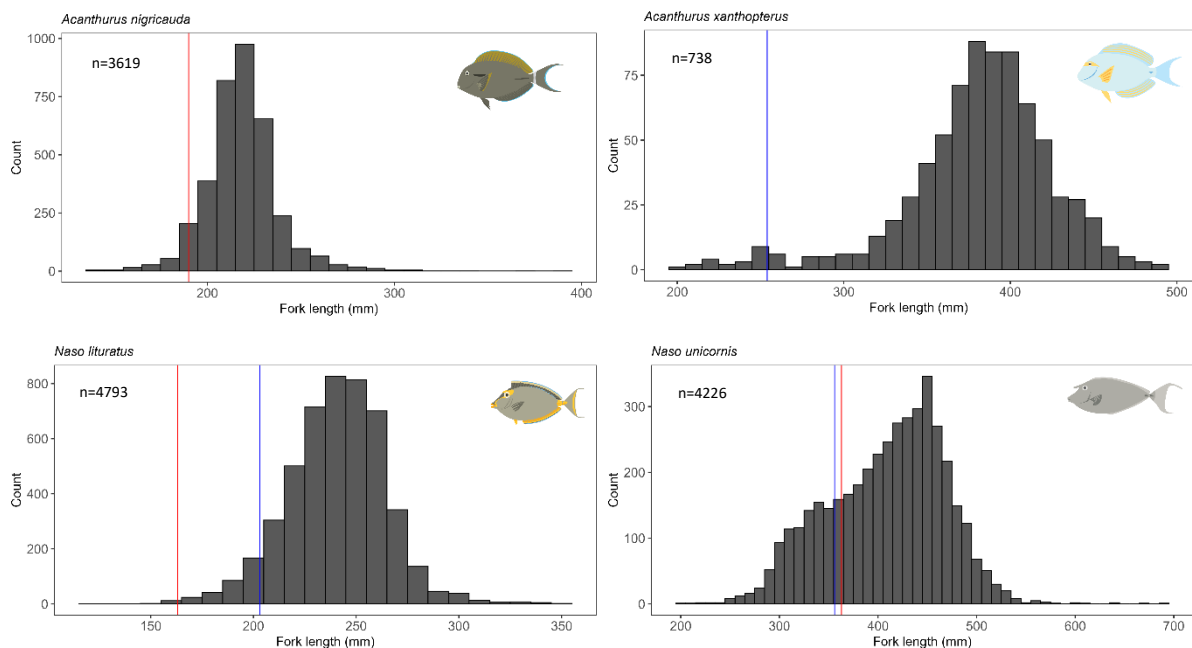


Figure 6. Length-frequency histograms for species in the Acanthuridae family with >100 length measurements from 2014 to 2024. Red lines indicate locally-derived size at maturity estimates (L₅₀) and blue lines indicate proposed minimum size limits from the KSLCFMP.

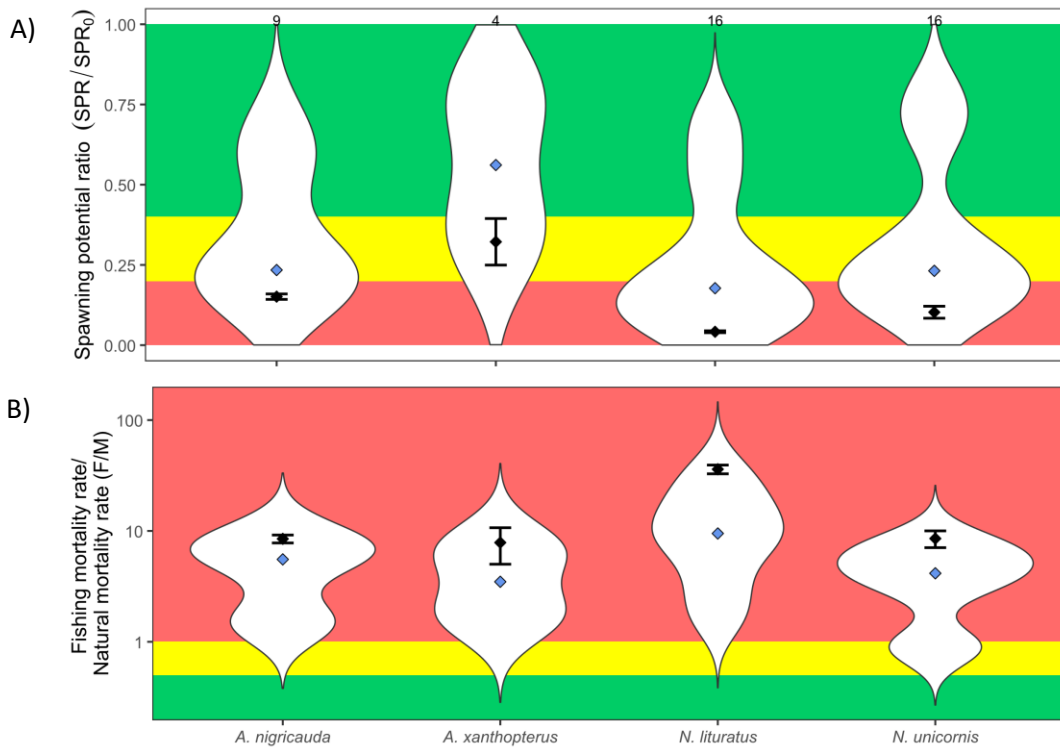


Figure 7. Violin plots showing estimated spawning potential ratio (SPR/SPR_0) (A) and fishing mortality rate/natural mortality rate (F/M) (B) for Acanthuridae species with >100 length measurements from 2014 to 2024. Data includes estimates from all scenarios run for each species, with the best scenarios shown by black points with error bars showing 95% confidence intervals, and blue points showing median values. The number of scenarios run for each species are shown at the top. Red areas indicate stocks are being overfished ($SPR < 0.2$ and $F/M > 1$), yellow areas indicate stocks that are vulnerable to being overfished ($SPR: 0.2 - 0.4$ and $F/M: 0.5 - 1$), and green areas indicate stocks that are being fished sustainably ($SPR > 0.4$ and $F/M < 0.5$).

Lethrinidae (emperors)

The percentage of immature fish landed from 2014 to 2024 was 3.4% for *Lethrinus obsoletus* (chudech), 32.6% for *Lethrinus olivaceus* (melangmud), 9.8% for *Lethrinus xanthochilus* (mechur), and 23.5% for *Monotaxis grandoculis* (besechamel) (Figure 8). There are no locally derived size of maturity estimates for *Lethrinus erythracanthus* (menges) and *Lethrinus lentjan* (metengui). Three out of six assessed Lethrinidae species had best SPR estimates below 0.2, including *L. obsoletus* – 0.08 (0.07-0.10 95% CI), *L. olivaceus* – 0.13 (0.10-0.16 95% CI), and *L. xanthochilus* – 0.11 (0.09-0.12 95% CI). The rest of the species had SPR estimates at or above 0.2, including *L. erythracanthus* – 0.31 (0.17-0.44 95% CI), *L. lentjan* – 0.33 (0.27-0.39 95% CI), and *M. grandoculis* – 0.20 (0.13-0.27 95% CI) (Figure 9, Appendix 5). All Lethrinidae species had an F/M above one.

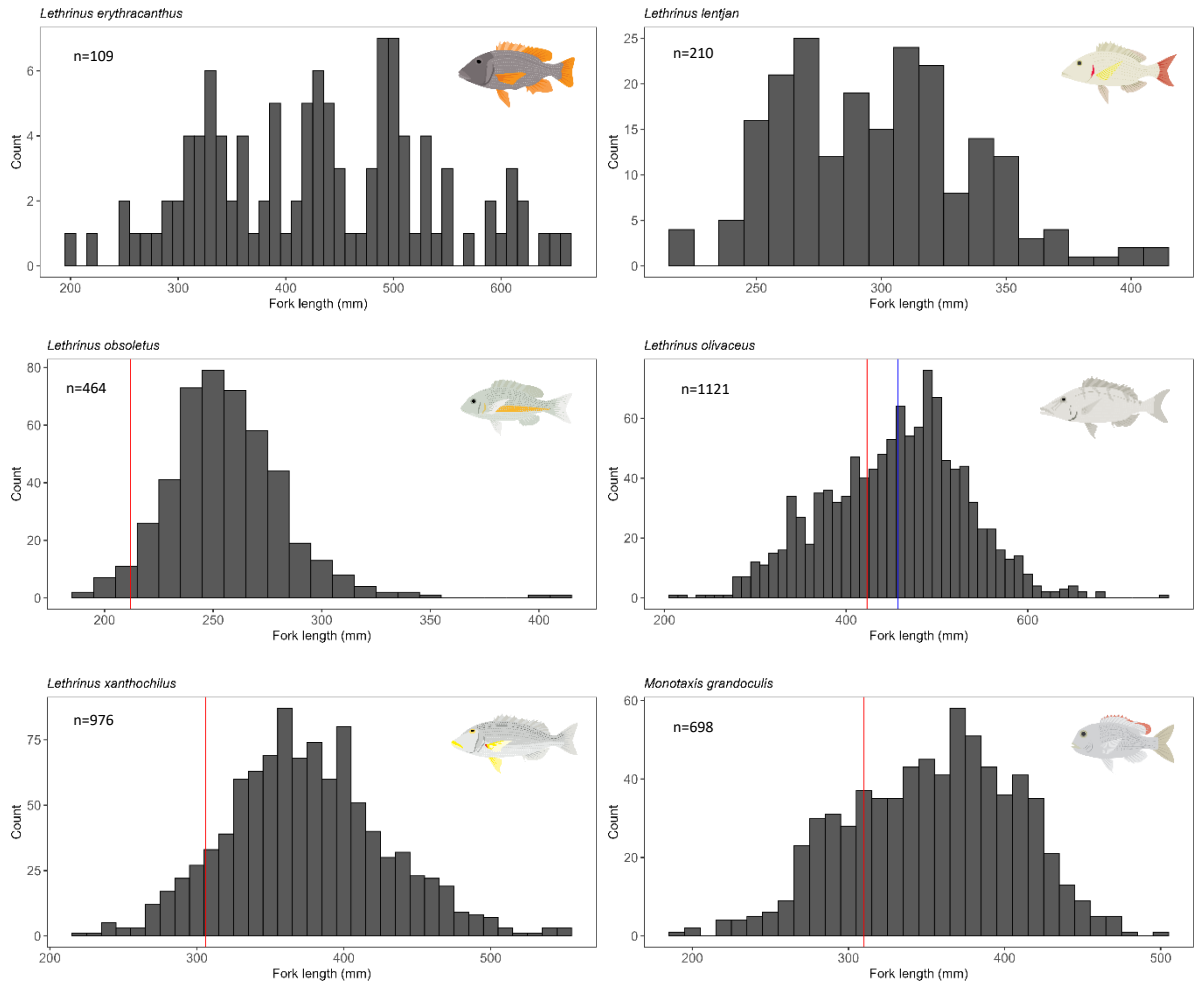


Figure 8. Length-frequency histograms for species in the Lethrinidae family with >100 length measurements from 2014 to 2024. Red lines indicate locally-derived size at maturity estimates (L_{50}) and blue lines indicate proposed minimum size limits from the KSLCFMP.

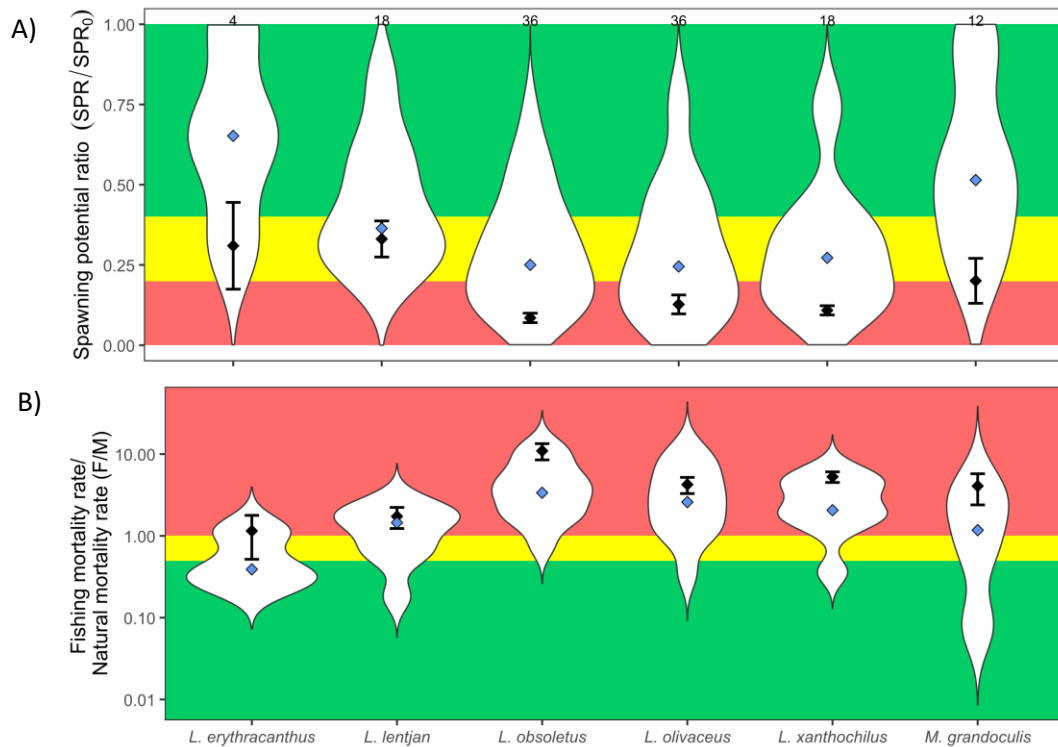


Figure 9. Violin plots showing estimated spawning potential ratio (SPR/SPR_0) (A) and fishing mortality rate/natural mortality rate (F/M) (B) for Lethrinidae species with >100 length measurements from 2014 to 2024. Data includes estimates from all scenarios run for each species, with the best scenarios shown by black points with error bars showing 95% confidence intervals, and blue points showing median values. The number of scenarios run for each species are shown at the top. Red areas indicate stocks are being overfished ($SPR < 0.2$ and $F/M > 1$), yellow areas indicate stocks that are vulnerable to being overfished ($SPR: 0.2 - 0.4$ and $F/M: 0.5 - 1$), and green areas indicate stocks that are being fished sustainably ($SPR > 0.4$ and $F/M < 0.5$).

Lutjanidae (snappers)

The percentage of immature fish landed from 2014 to 2024 was 35.9% for *Lutjanus bohar* (kedesau), and 16.5% for *L. gibbus* (Figure 10). There are no locally derived size of maturity estimates for *Lutjanus monostigma* (kesebii) and *Lutjanus vitta* (dodes). Two out of four assessed Lutjanidae species had best SPR estimates below 0.2, including *L. bohar* – 0.07 (0.05-0.08 95% CI) and *L. gibbus* – 0.05 (0.05-0.06 95% CI). The other two species had SPR estimates above 0.2, including *L. monostigma* – 0.41 (0.31-0.50 95% CI) and *L. vitta* – 0.38 (0.32-0.45 95% CI) (Figure 11, Appendix 5). All Lutjanidae species had an F/M above one.

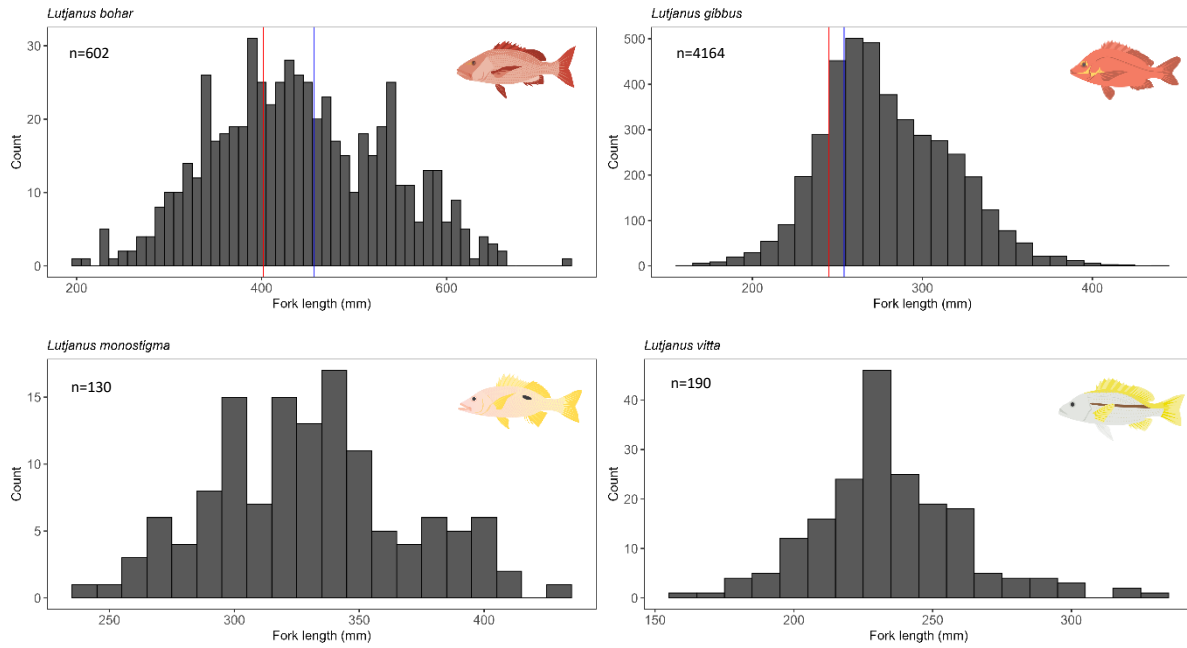


Figure 10. Length-frequency histograms for species in the Lutjanidae family with >100 length measurements from 2014 to 2024. Red lines indicate locally-derived size at maturity estimates (L_{50}) and blue lines indicate proposed minimum size limits from the KSLCFMP.

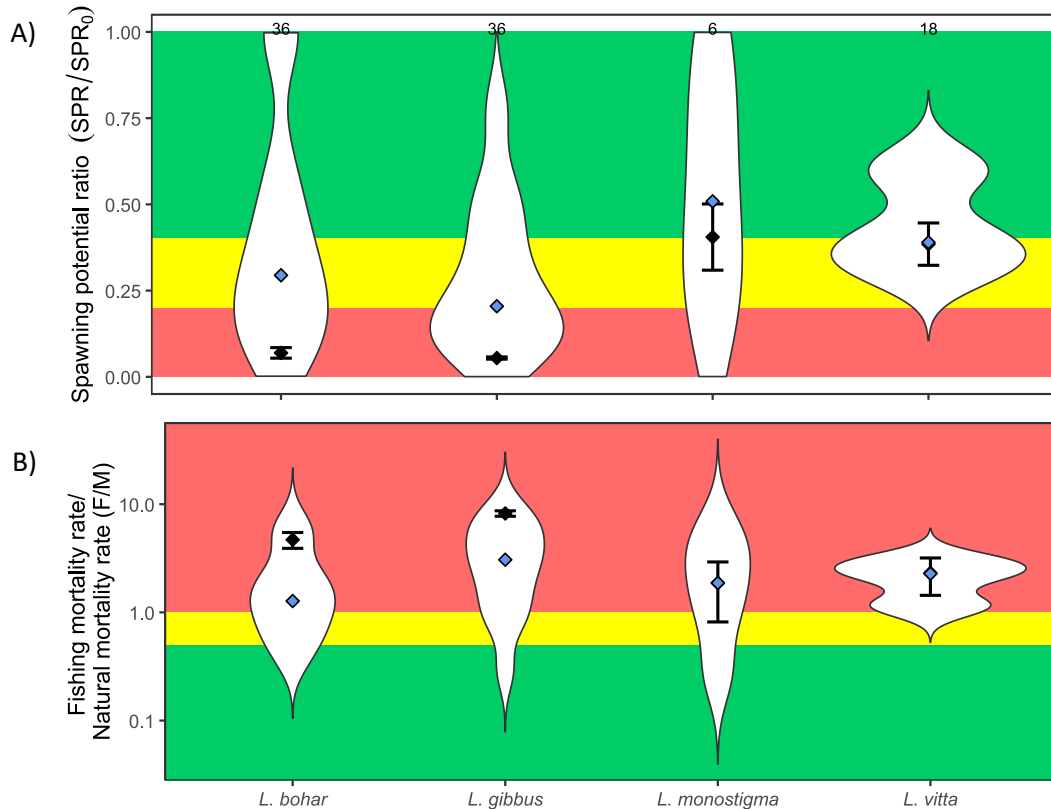


Figure 11. Violin plots showing estimated spawning potential ratio (SPR/SPR_0) (A) and fishing mortality rate/natural mortality rate (F/M) (B) for Lutjanidae species with >100 length measurements from 2014 to 2024. Data includes estimates from all scenarios run for each species, with the best scenarios shown by black points with error bars showing 95% confidence intervals, and blue points showing median values. The number of scenarios run for each species are shown at the top. Red areas indicate stocks are being overfished ($SPR < 0.2$ and $F/M > 1$), yellow areas indicate stocks that are vulnerable to being overfished ($SPR: 0.2 - 0.4$ and $F/M: 0.5 - 1$), and green areas indicate stocks that are being fished sustainably ($SPR > 0.4$ and $F/M < 0.5$).

Serranidae (groupers)

The percentage of immature fish landed from 2014 to 2024 was 37.9% for *E. polyphkadion*, 28.2% for *Plectropomus areolatus* (tiau), 11.9% for *Plectropomus leopardus* (red tiau), and 12.6% for *Variola louti* (baslokil) (Figure 12). There are no locally derived size of maturity estimates for *Aethaloperca rogaa* (chubei) or *Cephalopholis argus* (mengardechelucheb. Only *E. polyphkadion* had a best SPR estimate below 0.2 of 0.11 (0.08-0.14 95% CI). The rest of the species had SPR estimates above 0.2, including *A. rogaa* – 0.48 (0.40-0.57 95% CI), *C. argus* – 0.48 (0.44-0.53 95% CI), *P. areolatus* – 0.30 (0.20-0.40 95% CI), *P. leopardus* – 0.21 (0.19-0.24 95% CI), and *V. louti* – 0.25 (0.11-0.39 95% CI) (Figure 13, Appendix 5). All

Serranidae species had an F/M above one. The current national minimum size limit of 12 inches (304.8 mm) for five species of grouper is below the size of maturity for *E. polyphemus* and *P. areolatus* (Figure 12).

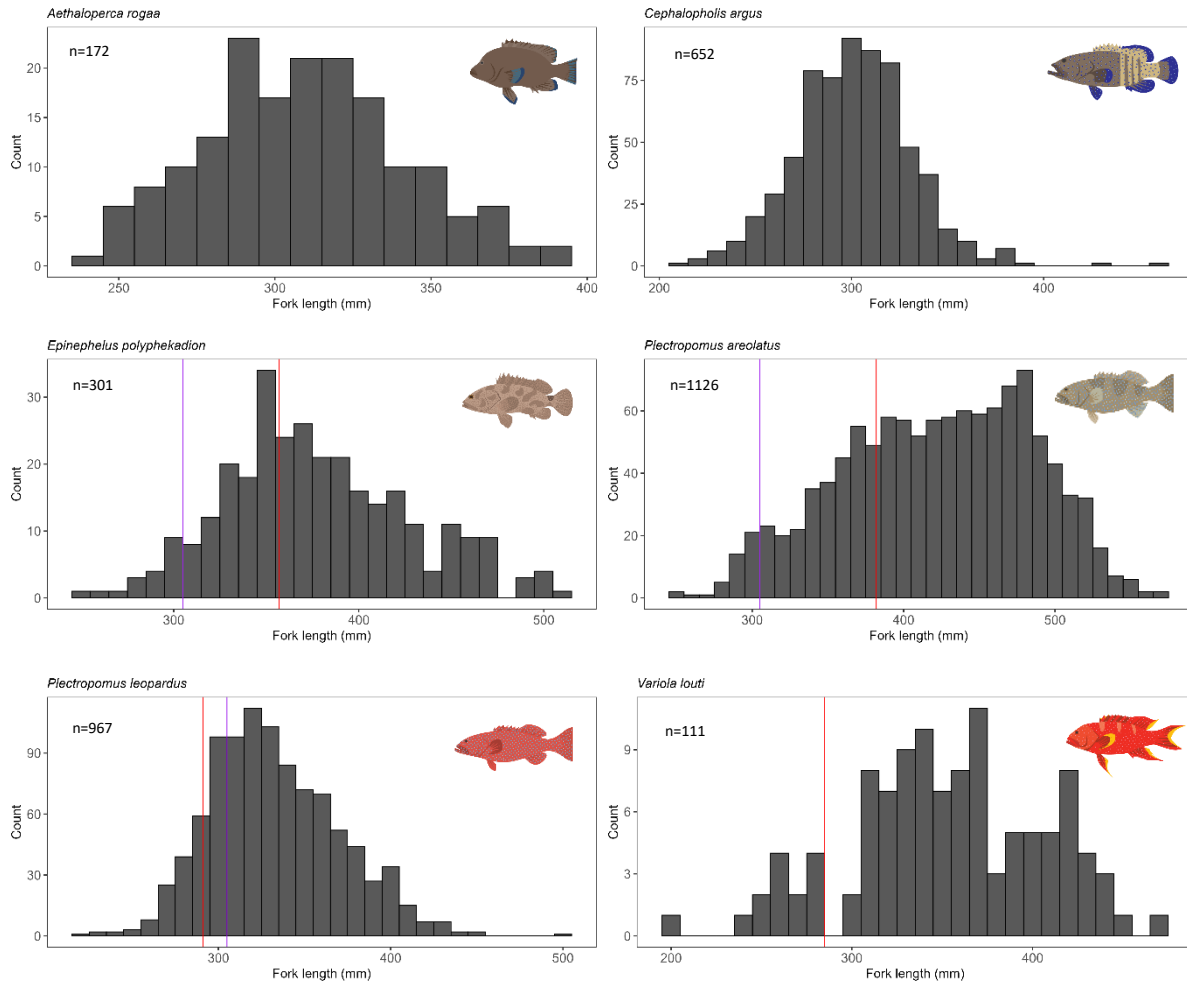


Figure 12. Length-frequency histograms for species in the Serranidae family with >100 length measurements from 2014 to 2024. Red lines indicate locally-derived size at maturity estimates (L_{50}) and purple lines indicate current national minimum size limits.

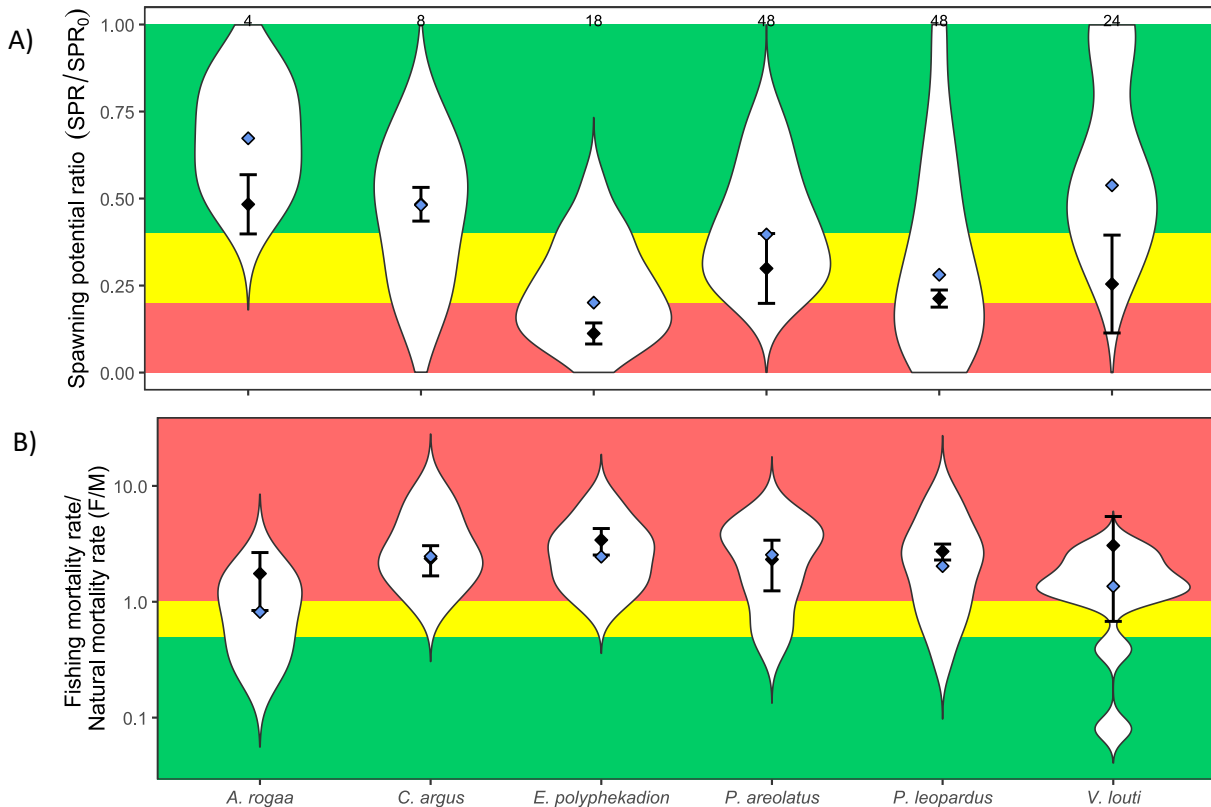
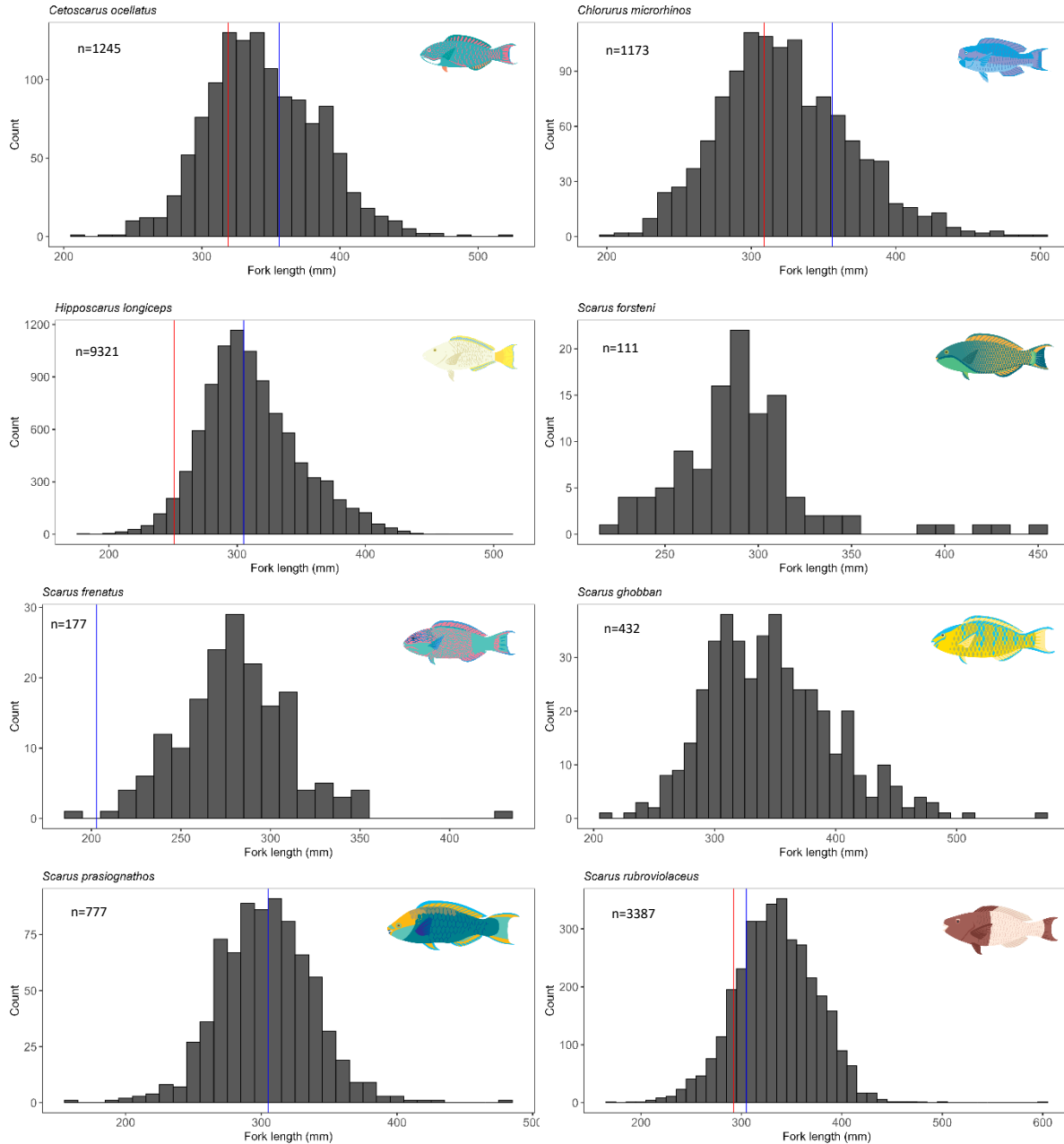


Figure 13. Violin plots showing estimated spawning potential ratio (SPR/SPR_0) (A) and fishing mortality rate/natural mortality rate (F/M) (B) for Serranidae species with >100 length measurements from 2014 to 2024. Data includes estimates from all scenarios run for each species, with the best scenarios shown by black points with error bars showing 95% confidence intervals, and blue points showing median values. The number of scenarios run for each species are shown at the top. Red areas indicate stocks are being overfished ($SPR < 0.2$ and $F/M > 1$), yellow areas indicate stocks that are vulnerable to being overfished ($SPR: 0.2 - 0.4$ and $F/M: 0.5 - 1$), and green areas indicate stocks that are being fished sustainably ($SPR > 0.4$ and $F/M < 0.5$).

Scaridae (parrotfish)

The percentage of immature fish landed from 2014 to 2024 was 26.8% for *Cetoscarus ocellatus* (beyadel, ngesngis), 39.8% for *Chlorurus microrhinos* (otord), 3.4% for *H. longiceps*, and 13.4% for *Scarus rubroviolaceus* (mesekelat mellemau) (Figure 14). There are no locally derived size of maturity estimates for *Scarus forsteni* (mul), *Scarus frenatus*, *Scarus ghobban* (mertebetabek), *Scarus prasiognathos* (melechotech a chau), and *Scarus schlegeli*. Five out of nine assessed Scaridae species had best SPR estimates below 0.2, including *C. ocellatus* – 0.10 (0.09-0.11 95% CI), *C. microrhinos* – 0.06 (0.05-0.07 95% CI), *H. longiceps* – 0.16 (0.15-0.16 95%

CI), *S. ghobban* – 0.18 (0.16-0.21 95% CI), and *S. rubroviolaceus* – 0.08 (0.07-0.08 95% CI). The rest of the species had SPR estimates above 0.2, including *S. forsteni* – 0.30 (0.23-0.37 95% CI), *S. frenatus* – 0.50 (0.40-0.61 95% CI), *S. prasiognathos* – 0.39 (0.35-0.43 95% CI), and *S. schlegeli* – 0.56 (0.46-0.66 95% CI) (Figure 15, Appendix 5). All Scaridae species had an F/M above one.



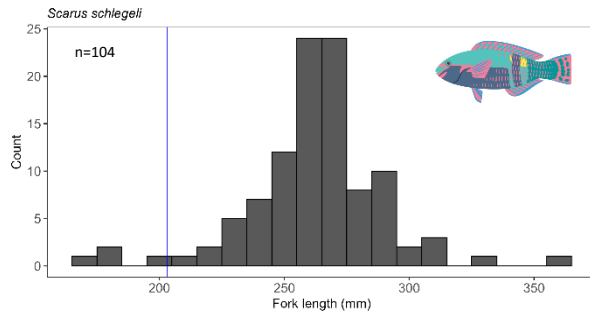


Figure 14. Length-frequency histograms for species in the Scaridae family with >100 length measurements from 2014 to 2024. Red lines indicate locally-derived size at maturity estimates (L_{50}) and blue lines indicate proposed minimum size limits from the KSLCFMP.

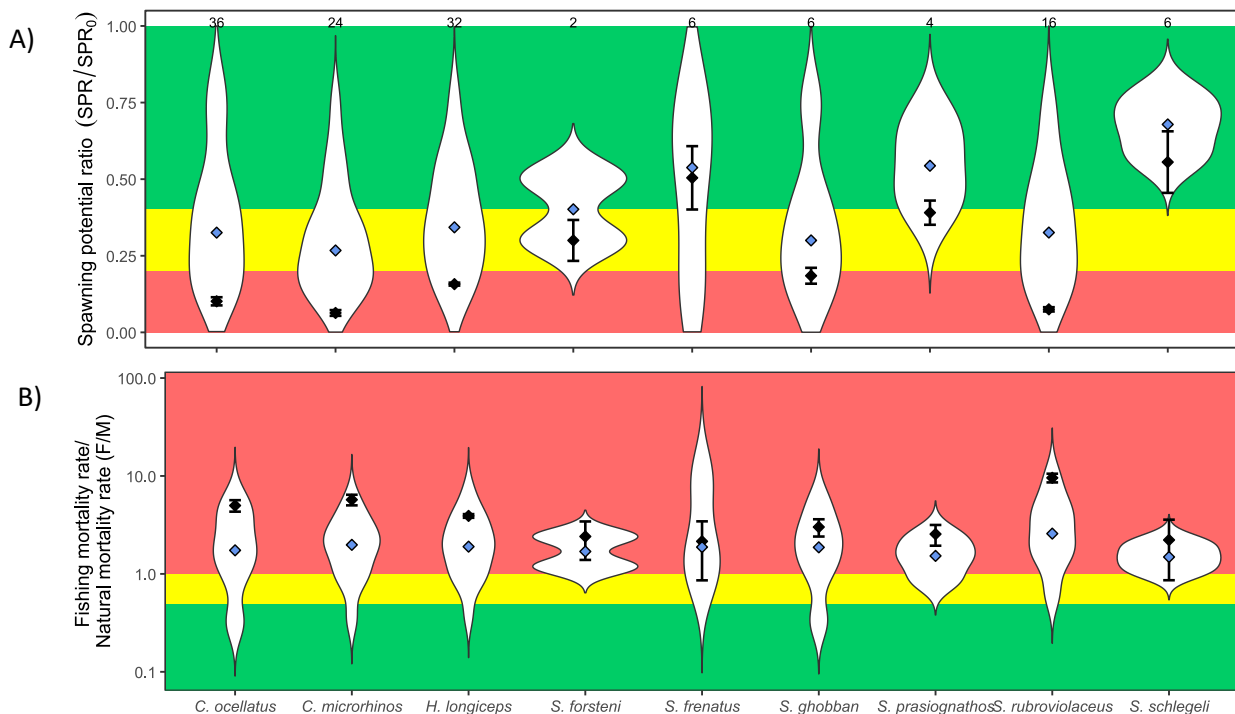
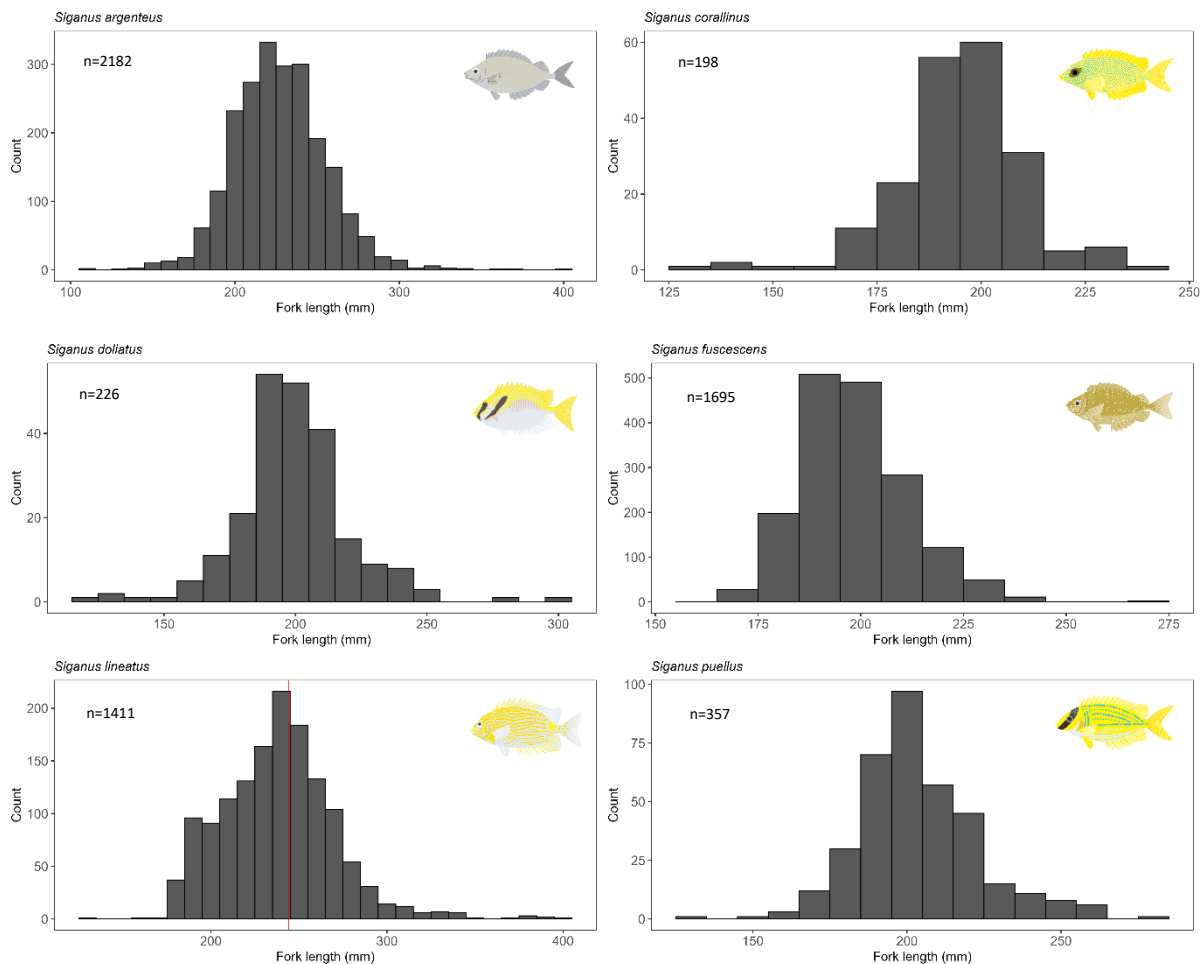


Figure 15. Violin plots showing estimated spawning potential ratio (SPR/SPR_0) (A) and fishing mortality rate/natural mortality rate (F/M) (B) for Scaridae species with >100 length measurements from 2014 to 2024. Data includes estimates from all scenarios run for each species, with the best scenarios shown by black points with error bars showing 95% confidence intervals, and blue points showing median values. The number of scenarios run for each species are shown at the top. Red areas indicate stocks are being overfished ($SPR < 0.2$ and $F/M > 1$), yellow areas indicate stocks that are vulnerable to being overfished ($SPR: 0.2 - 0.4$ and $F/M: 0.5 - 1$), and green areas indicate stocks that are being fished sustainably ($SPR > 0.4$ and $F/M < 0.5$).

Siganidae (rabbitfish)

The percentage of immature fish landed from 2014 to 2024 was 58.1% for *Siganus lineatus* (kelsebuul) and 35.6% for *Siganus punctatus* (bebael) (Figure 16). There are no locally derived size of maturity estimates for *Siganus argenteus* (beduut), *Siganus corallinus* (reked), *Siganus doliatus* (reked), *S. fuscescens*, and *Siganus puellus* (reked). All seven assessed Siganiidae species had best SPR estimates above 0.2, including *S. argenteus* – 0.62 (0.59-0.65 95% CI), *S. corallinus* – 0.76 (0.67-0.85 95% CI), *S. doliatus* – 0.81 (0.74-0.88 95% CI), *S. fuscescens* – 0.70 (0.69-0.72 95% CI), *S. lineatus* – 0.36 (0.29-0.43 95% CI), *S. puellus* – 0.65 (0.60-0.69 95% CI), and *S. punctatus* – 0.41 (0.38-0.44 95% CI) (Figure 17, Appendix 5). Five species had an F/M above one (*S. argenteus*, *S. corallinus*, *S. fuscescens*, *S. puellus* and *S. punctatus*), and two species had an F/M below one (*S. doliatus* and *S. lineatus*).



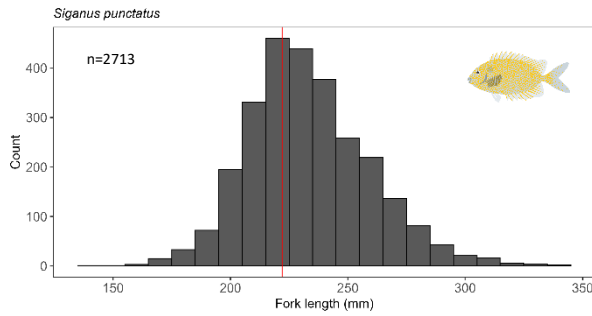


Figure 16. Length-frequency histograms for species in the Siganidae family with >100 length measurements from 2014 to 2024. Red lines indicate locally-derived size at maturity estimates (L_{50}).

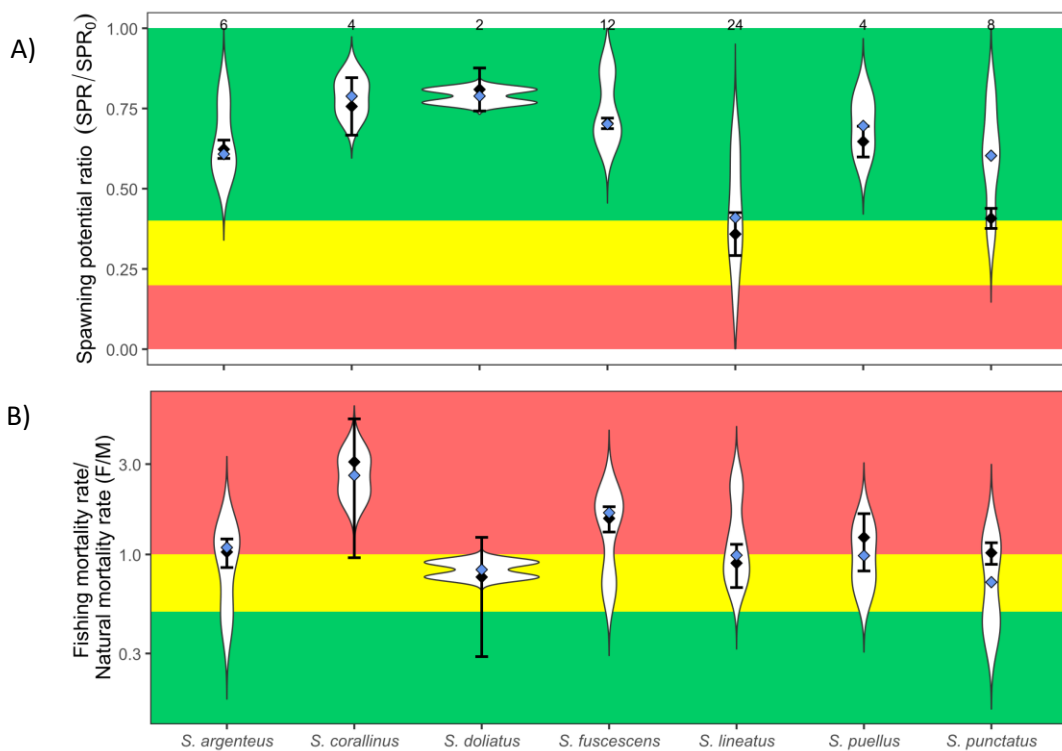


Figure 17. Violin plots showing estimated spawning potential ratio (SPR/SPR_0) (A) and fishing mortality rate/natural mortality rate (F/M) (B) for Siganidae species with >100 length measurements from 2014 to 2024. Data includes estimates from all scenarios run for each species, with the best scenarios shown by black points with error bars showing 95% confidence intervals, and blue points showing median values. The number of scenarios run for each species are shown at the top. Red areas indicate stocks are being overfished ($SPR < 0.2$ and $F/M > 1$), yellow areas indicate stocks that are vulnerable to being overfished ($SPR: 0.2 - 0.4$ and $F/M: 0.5 - 1$), and green areas indicate stocks that are being fished sustainably ($SPR > 0.4$ and $F/M < 0.5$).

Carangidae (trevallies)

The percentage of immature fish landed from 2014 to 2024 was 1.6% for *Caranx melampygus* (orwidel) (Figure 18). There are no locally derived size of maturity estimates for *Atule mate* (klspeached) and *Ferdauia orthogrammus* (otewot). All three assessed Carangidae species had best SPR estimates above 0.2, including *A. mate* – 0.80 (0.72-0.88 95% CI), *C. melampygus* – 0.56 (0.39-0.72 95% CI), and *F. orthogrammus* – 0.57 (0.43-0.70 95% CI) (Figure 19, Appendix 5). Two species had an F/M above one, including *A. mate* and *F. orthogrammus*, whereas *C. melampygus* had an F/M value below one.

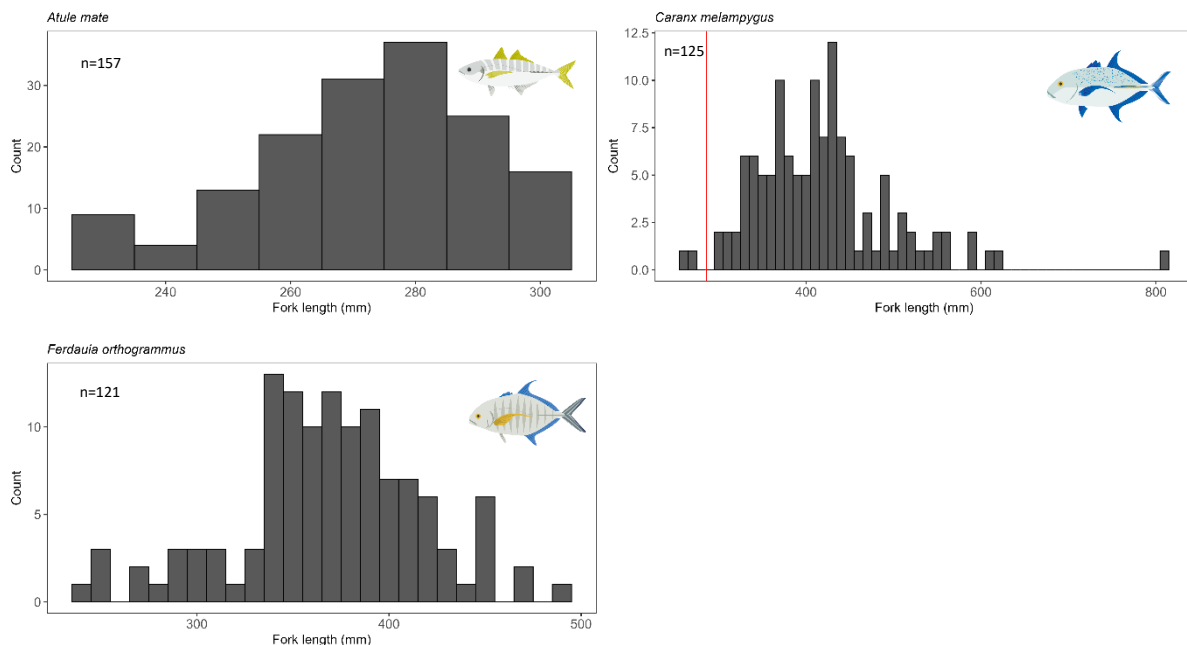


Figure 18. Length-frequency histograms for species in the Carangidae family with >100 length measurements from 2014 to 2024. Red lines indicate locally-derived size at maturity estimates (L₅₀).

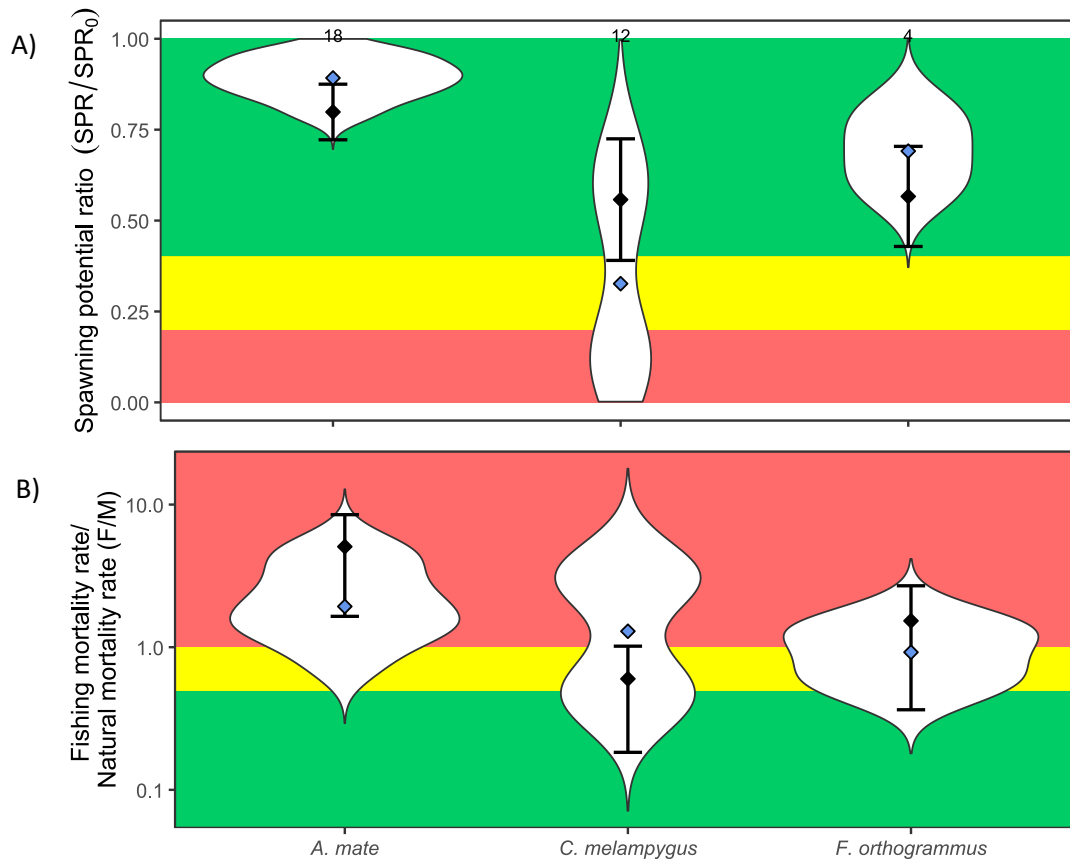


Figure 19. Violin plots showing estimated spawning potential ratio (SPR/SPR_0) (A) and fishing mortality rate/natural mortality rate (F/M) (B) for Carangidae species with >100 length measurements from 2014 to 2024. Data includes estimates from all scenarios run for each species, with the best scenarios shown by black points with error bars showing 95% confidence intervals, and blue points showing median values. The number of scenarios run for each species are shown at the top. Red areas indicate stocks are being overfished ($SPR < 0.2$ and $F/M > 1$), yellow areas indicate stocks that are vulnerable to being overfished ($SPR: 0.2 - 0.4$ and $F/M: 0.5 - 1$), and green areas indicate stocks that are being fished sustainably ($SPR > 0.4$ and $F/M < 0.5$).

Mullidae (goatfish)

The percentage of immature fish landed from 2014 to 2024 was 4.1% for *P. barberinus* (Figure 20). There are no locally derived size of maturity estimates for *P. cyclostomus*. Both species had best SPR estimates above 0.2, including *P. barberinus* – 0.73 (0.70-0.76 95% CI) and *P. cyclostomus* – 0.85 (0.78-0.92 95% CI) (Figure 21, Appendix 5). Both species also had F/M values below one.

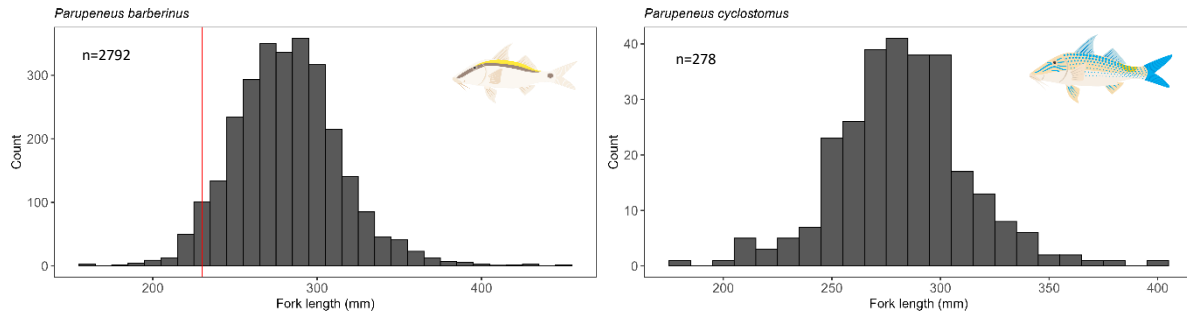


Figure 20. Length-frequency histograms for species in the Mullidae family with >100 length measurements from 2014 to 2024. Red lines indicate locally-derived size at maturity estimates (L_{50}).

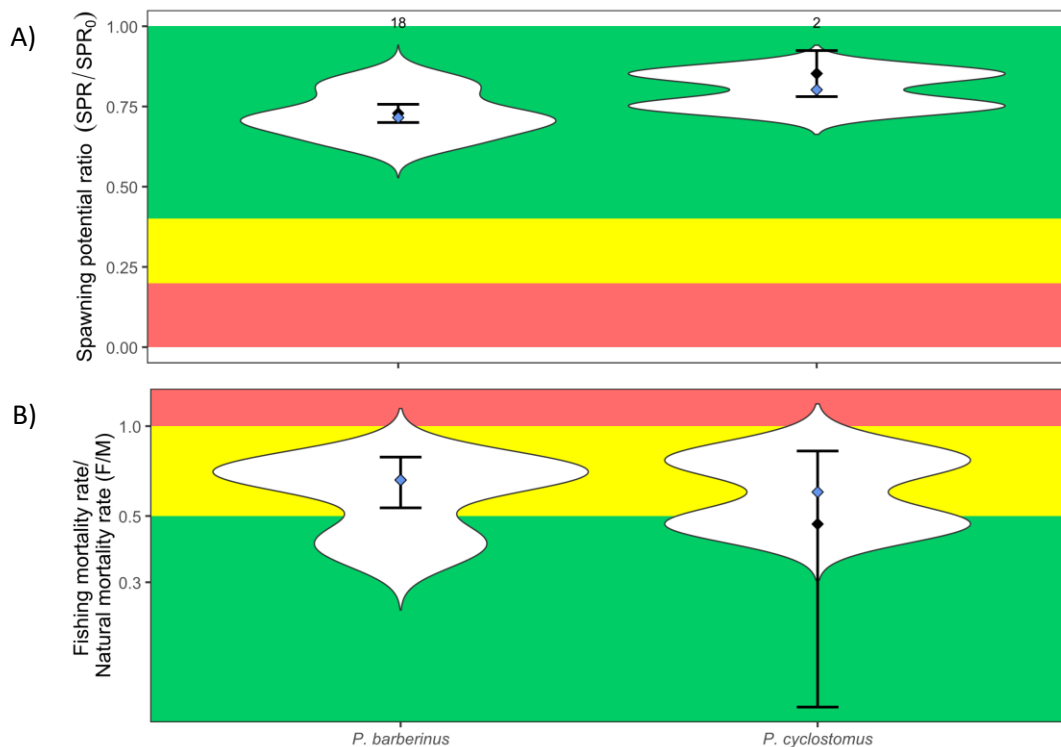


Figure 21. Violin plots showing estimated spawning potential ratio (SPR/SPR_0) (A) and fishing mortality rate/natural mortality rate (F/M) (B) for Mullidae species with >100 length measurements from 2014 to 2024. Data includes estimates from all scenarios run for each species, with the best scenarios shown by black points with error bars showing 95% confidence intervals, and blue points showing median values. The number of scenarios run for each species are shown at the top. Red areas indicate stocks are being overfished ($SPR < 0.2$ and $F/M > 1$), yellow areas indicate stocks that are vulnerable to being overfished ($SPR: 0.2 - 0.4$ and $F/M: 0.5 - 1$), and green areas indicate stocks that are being fished sustainably ($SPR > 0.4$ and $F/M < 0.5$).

Haemulidae (sweetlips)

The percentage of immature fish landed from 2014 to 2024 was 22.0% for *Plectrohinchus lineatus* (yaus) (Figure 22). There are no locally derived size of maturity estimates for *Plectrohinchus albovittatus* (biki). Both species had best SPR estimates above 0.2, including *P. albovittatus* - 0.51 (0.36-0.66 95% CI), and *P. lineatus* - 0.27 (0.21-0.33 95% CI) (Figure 23, Appendix 5). *P. albovittatus* also had an F/M value below one, whereas *P. lineatus* had an F/M value above one.

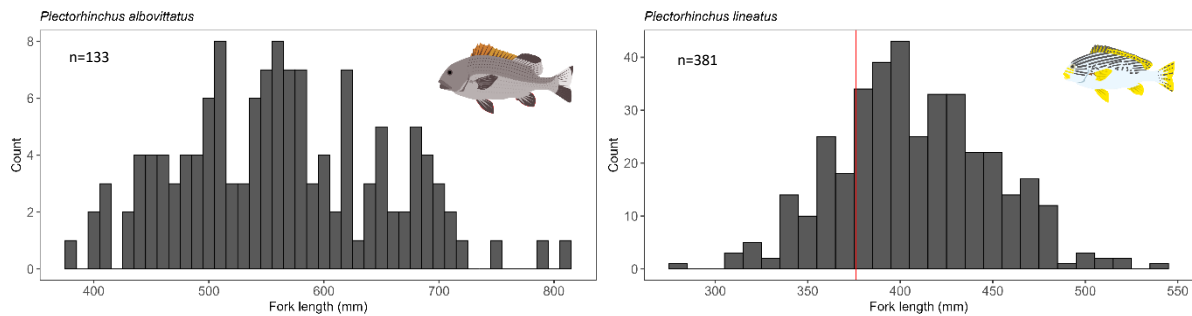


Figure 22. Length-frequency histograms for species in the Haemulidae family with >100 length measurements from 2014 to 2024. Red lines indicate locally-derived size at maturity estimates (L₅₀).

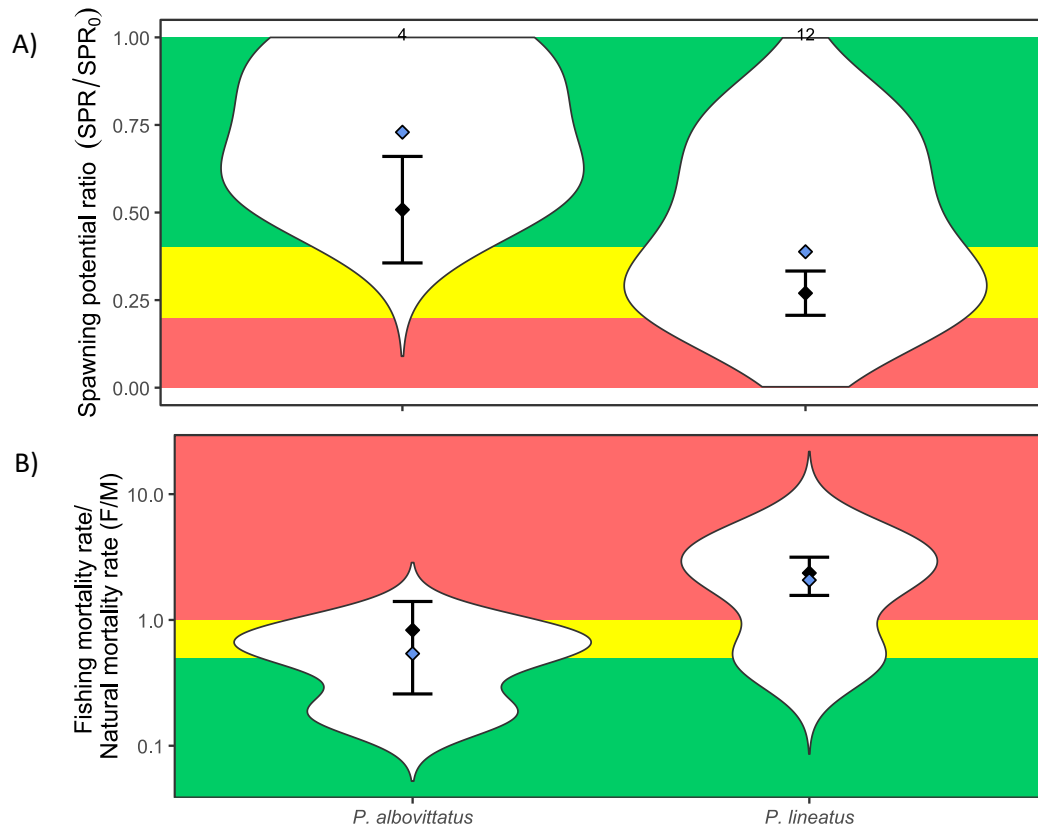


Figure 23. Violin plots showing estimated spawning potential ratio (SPR/SPR₀) (A) and fishing mortality rate/natural mortality rate (F/M) (B) for Haemulidae species with >100 length measurements from 2014 to 2024. Data includes estimates from all

scenarios run for each species, with the best scenarios shown by black points with error bars showing 95% confidence intervals, and blue points showing median values. The number of scenarios run for each species are shown at the top. Red areas indicate stocks are being overfished ($SPR < 0.2$ and $F/M > 1$), yellow areas indicate stocks that are vulnerable to being overfished ($SPR: 0.2 - 0.4$ and $F/M: 0.5 - 1$), and green areas indicate stocks that are being fished sustainably ($SPR > 0.4$ and $F/M < 0.5$).

Scombridae (mackerel) and Holocentridae (squirrelfish)

There are no locally derived size of maturity estimates for *Rastrelliger kanagurta* (smach) and *Sargocentron spiniferum* (desachel) (Figure 24). Both species had best SPR estimates above 0.2, including *R. kanagurta* – 0.60 (0.53-0.66) and *S. spiniferum* – 0.40 (0.38-0.42) (Figure 25, Appendix 5). Both species also had F/M values above one.

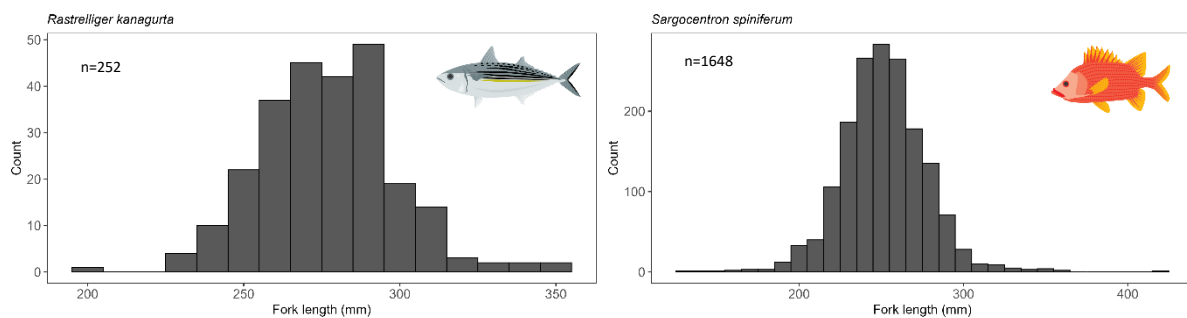


Figure 24. Length-frequency histograms for species in the Scombridae and Holocentridae families with >100 length measurements from 2014 to 2024.

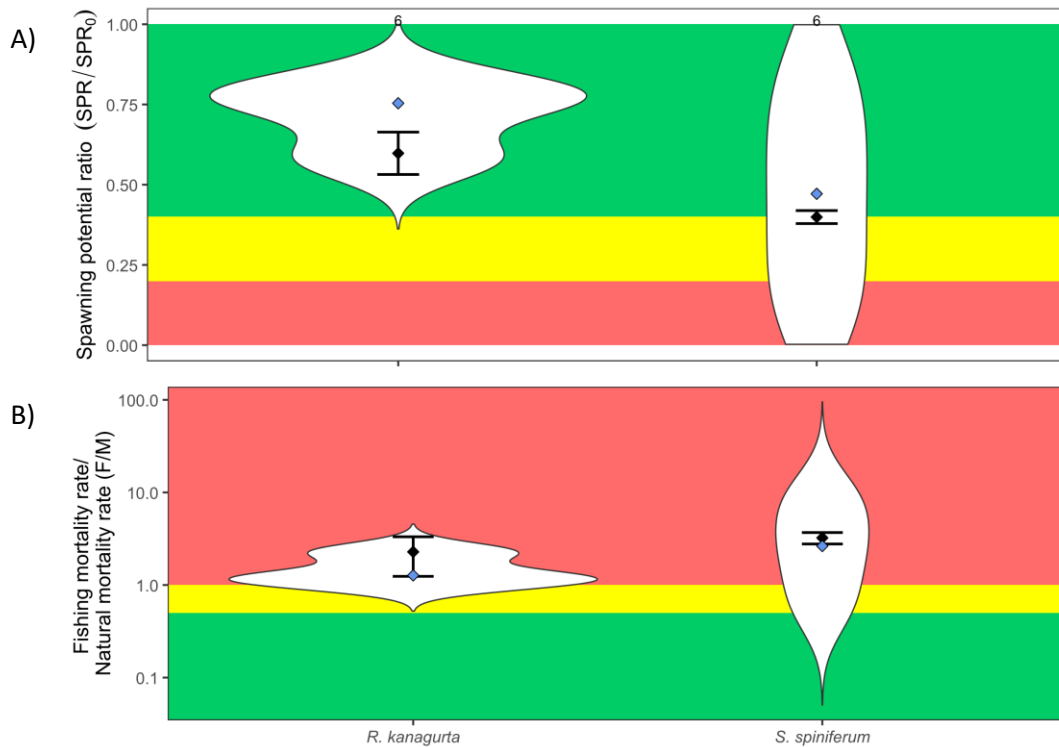


Figure 25. Violin plots showing estimated spawning potential ratio (SPR/SPR_0) (A) and fishing mortality rate/natural mortality rate (F/M) (B) for Scombridae and Holocentridae species with >100 length measurements from 2014 to 2024. Data includes estimates from all scenarios run for each species, with the best scenarios shown by black points with error bars showing 95% confidence intervals, and blue points showing median values. The number of scenarios run for each species are shown at the top. Red areas indicate stocks are being overfished ($SPR < 0.2$ and $F/M > 1$), yellow areas indicate stocks that are vulnerable to being overfished ($SPR: 0.2 - 0.4$ and $F/M: 0.5 - 1$), and green areas indicate stocks that are being fished sustainably ($SPR > 0.4$ and $F/M < 0.5$).

4. Discussion

4.1 Fisher interviews and species composition

This report aimed to assess the status of the top reef fishery species in Palau using fish landing data collected over the last 40 years. Fisher interviews conducted in 2017 and 2023 showed that fish being sold at the JR5 fish market mostly originate from Koror state waters and, to a lesser extent, other states. In 2023, fishing mostly took place in the southwestern reefs and the highest percentage of fishing trips in 2017 and 2023 occurred in the outer reef and back reef habitats. Commercially important reef fish biomass has previously been found to be the highest in western outer reefs and lowest in the inner reefs of Palau, based on PICRC's fishery-independent surveys (Muller Karanassos et al., 2023). The higher productivity of the southwestern outer reefs is likely why this is a fishing hot spot and perhaps fishers choose to use the back

reef habitat during periods of bad weather, although the relationship with weather conditions was not explored.

Night spearfishing was the predominant fishing method used in 2017 and 2023, with a higher occurrence of handline fishing in 2017 compared to 2023. This is because surveys were conducted in both the morning and afternoon in 2017 whereas in 2023 the survey was only conducted in the morning when night spear fishers were returning, due to logistical limitations. Night spearfishing is known to be a very effective fishing method (Gillett and Moy, 2006) and has even been banned in some countries due to its ability to reduce fish populations to low levels (Harding et al., 2022). Interestingly, CPUE for night spearfishing increased from 2017 to 2023, which may indirectly indicate recovery of fish stocks over time; however catch rates can be affected by many factors such as weather conditions, fishing locations, tide and moon phase, seasons, etc. and therefore CPUE data should be treated with caution (Maunder et al., 2006). In 2020, the ban on exporting reef fish was introduced and COVID-19 led to a rapid decline in tourism. These factors would have led to a reduction in demand for reef fish and may have caused a decrease in the number of fishing trips, although no official data is available to support this. In addition, the 10% Palau Goods and Services Tax (PGST) implemented in January 2023 led to an increase in fuel prices, which may have further reduced the number of boats going out. These combined factors may have reduced the pressure on reef fish populations leading to recovery over time, however the time it takes for fish populations to recover can vary substantially based on the previous level of fishing pressure and species life histories. Some studies have shown it can take as little as three years for fish populations to recover (Halpern and Warner, 2002), whereas others predicted it would take up to 59 years of protection for heavily fished sites to recover (MacNeil et al., 2015). Furthermore, large predatory species, which are long-lived will take a longer time to recover compared to short-lived species (Russ and Alcala, 2004). It is not known whether the observed increase in CPUE was due to the recovery of small fast-growing species and with tourism increasing after COVID-19 it is likely this will increase the pressure on reef fish again.

The composition of reef fish landings has changed substantially over time, with a higher diversity of fish being landed from 2015 to 2023 compared to 1984 and 1991. Before 1994, there were no official fisheries regulations in Palau (apart from the

establishment of the Ngerukeuid MPA in 1956), which explains the high proportion of *E. polyphkadion* landed in 1984 and *S. fuscescens* in 1991, before the seasonal bans were introduced for these species through the Marine Protection Act in 1994. Furthermore, groupers were heavily targeted for exportation from Palau to Hong Kong for the live reef fish trade from the mid-1980s to 1999 (Graham, 2001). In addition, *Cheilinus undulatus* and *Bolbometopon muricatum* were still being landed in 1984 and 1991 before the complete ban on these species in 1994. Fishing pressure would have shifted from these species to others and together with the implementation of seasonal bans explains the increase in diversity of fish being landed from 2015-2023 compared to 1984 and 1991. Differences in fishing methods and timing of surveys would have also affected species composition, however this information was only collected in more recent surveys in 2017 and 2023.

4.2 LB-SPR assessments

Acanthuridae (surgeonfish/unicornfish)

Surgeonfish and unicornfish play a vital role in limiting the amount of macroalgae and turf algae on coral reefs through their role as grazers (Green and Bellwood, 2009). Several Acanthuridae species are commonly sold at the JR5 fish market, with *N. unicornis*, *N. lituratus*, *A. xanthopterus*, and *A. nigricauda* being the most desirable to fishers (Bejarano et al., 2013a). Minimum size limits were proposed for three Acanthuridae species in the KSLCFMP, including *N. unicornis*, *N. lituratus*, and *A. xanthopterus* (Appendix 1). For *A. xanthopterus*, the proposed minimum size limit of 10 inches (254 mm) is above the L_{50} value estimated using empirical equations (221 mm), however without a locally derived size of maturity estimate based on gonad examination, it is not possible to determine if this size limit will effectively protect immature individuals.

There was no proposed minimum size limit for *A. nigricauda*, however, given the popularity of this species and depleted status estimated in this study ($SPR < 0.2$ and $F/M > 1$), it is recommended that a minimum size limit of 209 mm or 228 mm is introduced to allow the species to recover. The size of maturity estimated for this species by Prince (2016b) was based on macroscopic observation of only five fish, and therefore the minimum size limit should be updated once more accurate life history information becomes available for this species in Palau.

N. unicornis is one of the most important commercial reef fish species in Micronesia (Houk et al., 2011), including Palau, with this species making up the highest percentage of landings by biomass in 2023 (16.8%). However, there have been concerns about overfishing of this species since the early 1990s, with fishers stating that stocks of *N. unicornis* were declining across the country (Johannes, 1991). Bejarano et al. (2013a) found that *N. unicornis* was the most vulnerable herbivore due to the heavy exploitation of this species and high proportion of immature individuals in fish landings. Furthermore, it is suspected that *N. unicornis* forms spawning aggregations, although this has not been confirmed. Fishers reported the formation of large groups when fish have eggs at both the full and new moons from February to August, making it easy for large numbers of fish to be caught on one fishing trip (Johannes et al., 1999, Sadovy, 2007). The low SPR and high F/M estimated in this study indicates this species has been depleted and a high proportion of immature fish are being landed. The size of maturity estimated by Prince (2016b) was only based on 14 samples, which highlights the need for improved life history information for this important species. In addition, the proposed minimum size limit of 356 mm is below the L_{50} (363 mm), and therefore a large proportion of immature fish will still be vulnerable to fishing. Given the popularity and vulnerability of this species in Palau, it is recommended that the proposed minimum size limit is increased to 399 mm or 436 mm as a more conservative approach, to ensure this species continues to be fished at a sustainable level and this should be updated once more accurate life history information becomes available (Appendix 1).

N. lituratus is highly desirable and heavily harvested in Palau (Bejarano et al., 2013a). This species has been observed to form aggregations of up to a few hundred fish on the last quarter moon from December to March at specific sites on the outer reefs of Palau (Johannes et al., 1999, Etpison and Colin, 2018). Although spawning has not been confirmed, the formation of aggregations by this species potentially makes them more vulnerable to overfishing (Sadovy and Domeier, 2005). The low SPR and high F/M estimated in this study indicates this species has been depleted, however most landed fish are mature. The proposed minimum size limit of 203 mm is above the L_{50} and should allow this species to recover over time.

Lethrinidae (emperors)

The KSLCFMP only proposed a minimum size limit for one emperor species, *L. olivaceus*, which is a popular reef fish species sold in restaurants in Palau. The low SPR and high F/M estimated in this study indicates this species has been depleted and a high proportion of immature fish are being landed. The proposed minimum size limit of 457 mm is lower than the estimated minimum size limit from our analysis and it is recommended that the minimum size limit is increased to 465 mm or 508 mm as a more conservative approach to allow this species to recover over time (Appendix 1). *L. olivaceus* is also suspected to form spawning aggregations around the new moon in Palau (Johannes, 1981, Sadovy, 2007), making it more vulnerable to overfishing. These aggregations are known to be targeted by fishers using handlines and overall catches of *L. olivaceus* are perceived to have declined over time (Sadovy, 2007, Johannes, 1981). PICRC has recently conducted additional surveys at a suspected spawning aggregation site and collected gonad samples which will provide more information to assist with the management of *L. olivaceus*.

L. xanthurus and *L. obsoletus* were also assessed to be depleted (low SPR and high F/M) and *M. grandoculis* was almost depleted (SPR=0.2) with a high percentage of immature fish being landed. These species have also been reported to form spawning aggregations in Palau (Johannes, 1981), although this has not been confirmed. Therefore, these species could also benefit from the implementation of minimum size limits (see Appendix 1). *L. erythracanthus* and *L. lentjan* were assessed as depleting but there is no locally derived size of maturity information for these species.

Lutjanidae (snappers)

The KSLCFMP proposed minimum size limits for two snapper species, *L. bohar* and *L. gibbus* (Appendix 1). *L. bohar* had a low SPR, high F/M, and a high percentage of landed fish were immature. *L. bohar* is also known to form spawning aggregations at several sites in Palau, with thousands of fish coming together to spawn (Sakaue et al., 2016), making them more vulnerable to overfishing. The proposed minimum size limit of 457.2 mm is above the L_{50} value and is similar to the size limit from our analysis (Appendix 1).

L. gibbus is one of the most heavily harvested snapper species in the Indo-Pacific region (Moore, 2019) including Palau, with high landings of this species recorded in Koror since 1984. There have been reports from fishers of fewer and smaller fish being caught in Koror State, causing concerns about the status of this species. This species is known to form large schools making it easy to catch them in large numbers using handlines and has been observed forming large spawning aggregations in French Polynesia (Nanninga and Spaet, 2017), although spawning aggregations have not been confirmed in Palau. *L. gibbus* had a low SPR and high F/M indicating this species has been depleted over time. The proposed minimum size limit of 254 mm is lower than our estimated size limit, and it is recommended that the proposed minimum size limit is increased to 270 mm or 294 mm as a more conservative approach (Appendix 1). The implementation of minimum size limits should help these two snapper species to recover over time and reduce the number of immature fish being landed. *L. monostigma* and *L. vitta* were assessed as depleting but there is no locally derived size of maturity information for these species.

Serranidae (groupers)

Several species of groupers are listed as vulnerable on the IUCN red list and have decreasing population trends worldwide, including *E. polyphkadion*, *P. areolatus*, and *E. fuscoguttatus*. These species are particularly vulnerable to overfishing due to their high economic value, late maturation, longevity, and formation of spawning aggregations at predictable times and locations (Sadovy de Mitcheson et al., 2020). In Palau, groupers have historically been targeted during their spawning aggregations, leading to the depletion of stocks and the disappearance of several aggregations over time (Johannes et al., 1999). It is for this reason that Palau introduced management regulations to protect five grouper species, including specific marine protected areas (Ebiil and Ngerumkaol), a seasonal ban during the peak spawning season (from April to October), and a national minimum size limit of 12 inches (304.8 mm) (Appendix 1). There were no minimum size limits proposed for groupers in the KSLCFMP, however from this survey, it is evident that the current minimum size limit is too low for most assessed species.

For *E. polyphkadion*, the minimum size limit is well below the L_{50} (357 mm) allowing many immature fish to still be caught. In addition, this species had a low SPR, high

F/M, and high percentage of immature fish landed, indicating this species has been overfished. This indicates that the current minimum size limit is too low and a revised minimum size limit of 393 mm, or 428 mm as a conservative approach, is implemented in Palau (including Koror state), to allow this species to recover. This is similar to the minimum size limit already in place for this species in Kayangel state (406 mm). For *P. areolatus*, although the best SPR estimate was 0.3, the F/M was high, and a high percentage of fish landed were immature. As with *E. polyphekadion*, the current national minimum size limit is well below the L_{50} (382 mm), allowing many immature fish to still be caught. A revised minimum size limit of at least 420 mm is recommended for Palau and Koror state, to allow this species to recover over time. Kayangel already has a minimum size limit in place for this species of 406 mm. Long-term monitoring at the Ebiil Channel spawning aggregation site showed an increase in the numbers of *P. areolatus* and *E. polyphekadion* from 2009 to 2019, indicating recovery over time (Sadovy de Mitcheson et al., 2020). Therefore, in addition to amending the minimum size limits, it is recommended that other known grouper spawning aggregation sites are protected from fishing activities. The protection of fish spawning aggregations that have previously been overfished can show recovery soon after implementation, with increases in recruitment, biomass, and size reported by several studies (e.g. Beets and Friedlander, 1999, Heppell et al., 2012). However, for conservation and fisheries benefits to be seen, it is important that these MPAs are well designed, properly enforced, and additional measures are implemented to control fishing mortality during non-aggregating times (Gruss et al., 2014). *P. leopardus* had low SPR of 0.21 and high F/M, indicating this species is likely overfished. The current size limit is slightly above the L_{50} but below our estimated size limit, and it is recommended that the proposed minimum size limit is increased to 320 mm or 349 mm as a more conservative approach (Appendix 1).

There was insufficient length data to conduct SPR analysis for *E. fuscoguttatus*, however the current minimum size limit is far below the size of maturity (563 mm) estimated for this species from Palau (Lindfield, 2020). Therefore it is recommended that the minimum size limit is increased to at least 619 mm, to protect this species from overfishing. There was also insufficient length data to conduct SPR analysis for *P. laevis* and there is no locally derived size of maturity estimate for this species in Palau. However, given the large size of this species, it is unlikely that the current

minimum size limit is effectively protecting immature fish. It is therefore recommended that size of maturity is estimated for this species through gonad examination and an appropriate minimum size limit is implemented based on the L_{50} value.

Although *V. louti* had an estimated SPR above 0.2, this was still less than 0.3, and F/M was high indicating depletion of this species. It is recommended that a minimum size limit of at least 314 mm is implemented to avoid the SPR decreasing further. *A. rogae* and *C. argus* were assessed as depleting but there is no locally derived size of maturity information for these species.

Scaridae (parrotfish)

Parrotfish are essential for sustaining the health of coral reef ecosystems through their crucial function of grazing on turf and macroalgae (Bonaldo et al., 2014). The top parrotfish species sold at the JR5 fish market are *H. longiceps*, *C. ocellatus*, *S. rubroviolaceus*, and *C. microrhinos*, which were all assessed to be vulnerable to overexploitation by Bejarano et al. (2013a). These large-bodied parrotfish have a particularly important role on coral reefs, since they can remove larger quantities of algae compared to smaller parrotfish species (Shantz et al., 2020). The KSLCFMP proposed minimum size limits for 16 parrotfish species (Appendix 1), however there was only sufficient data to assess nine of these species.

C. ocellatus, *C. microrhinos*, *H. longiceps*, *S. rubroviolaceus*, and *S. ghobban* were all assessed to be depleted, with low SPR and high F/M values. *C. ocellatus* and *C. microrhinos* also had high percentages of immature fish landed (27% and 40% respectively). The proposed minimum size limits for *C. ocellatus* and *C. microrhinos* are above the L_{50} values, however more conservative minimum size limits of 383 mm and 371 mm respectively (Appendix 1), may be more effective at reducing the number of immature fish being landed and to allow SPR to increase over time. The proposed minimum size limit for *S. rubroviolaceus* is just above the L_{50} and a higher minimum size limit of 321 mm or 350 mm is recommended to allow this species to recover over time.

H. longiceps is one of the top commercial reef fish species in Palau, as well as other Micronesian countries (Houk et al., 2011) and it has been the top parrotfish species recorded in fish landings since 1984 (Figure 4A). *H. longiceps* has been confirmed to form spawning aggregations at several sites on the outer reefs of Palau (Sadovy de

Mitcheson and Colin, 2012), potentially making this species more vulnerable to overfishing if they are targeted by fishers. The proposed minimum size limit of 305 mm is similar to our conservative estimated minimum size limit (Appendix 1), which should help this species to recover.

There was no proposed size limit for *S. ghobban* in the KSLCFMP and there is no locally derived size of maturity estimate for this species. Given the low SPR and high F/M estimated for this species, it is recommended that size of maturity is estimated through gonad examination and an appropriate minimum size limit is implemented based on the L_{50} value. There are also no locally derived size of maturity estimates for *S. forsteni*, *S. frenatus*, *S. prasiognathos*, and *S. schlegeli*, which were all assessed as depleting. Without locally derived size of maturity estimates based on gonad examination, it is not possible to accurately assess whether proposed minimum size limits will effectively protect immature fish.

Siganidae (rabbitfish)

Rabbitfish play a significant role in coral-algae phase shift reversals based on their algae diet (Green and Bellwood, 2009). There were no minimum size limits proposed in the KSLCFMP for rabbitfish, however species from this family are very important in Palau's reef fishery (Bejarano et al., 2013a). *S. lineatus* and *S. doliatus* were assessed as sustainable, however a very high percentage of landed *S. lineatus* were immature (58%). *S. lineatus* is farmed and restocked in Palau, which may be having a positive effect by reducing pressure on the natural population. *S. argenteus*, *S. corallinus*, *S. fuscescens*, *S. puellus*, and *S. punctatus* were assessed as depleting and a high percentage of landed *S. punctatus* were immature.

Rabbitfish are known to have a fast growth rate with a rapid population turnover and are generally not a concern for fisheries (Taylor et al., 2017). However, size of maturity information is only available for two rabbitfish species in Palau (*S. lineatus* and *S. punctatus*) and it is therefore recommended that size of maturity is estimated based on gonad examination for additional species including *S. fuscescens*, *S. argenteus*, *S. doliatus*, *S. puellus*, and *S. corallinus*. *S. fuscescens* (meyas) has historically been one of the most common reef fish species landed in Koror, and a seasonal ban was implemented to protect the spawning season from March 1st to May 31st in 1994 (amended to February 1st to March 31st in 2015). Without life history information, it is

difficult to assess how effective this management initiative has been, and whether the spawning population is increasing or decreasing over time. Size of maturity information will allow meyas and other species to be more accurately assessed in the future and for appropriate minimum size limits to be implemented based on L_{50} values if required.

Carangidae (trevallies)

No minimum size limits were proposed for species from the Carangidae family in the KSLCFMP. *C. melampygus* was assessed to be sustainable with a low percentage of immature fish landed, whereas *A. mate* and *F. orthogrammus* were assessed to be depleting. There are no locally derived size of maturity estimates for *A. mate* and *F. orthogrammus*. Carangidae species were mostly observed at the market in 2015 and were uncommon in the 2017 and 2023 surveys, therefore the collection of life history information for these species is not a priority.

Mullidae (goatfish)

No minimum size limits were proposed for species from the Mullidae family in the KSLCFMP. Goatfish (Mullidae) species are commonly seen at the market, with *P. barberinus* being one of the most popular. *P. barberinus* and *P. cyclostomus* were assessed to be sustainable, with a low percentage of immature *P. barberinus* landed. There is no locally derived size of maturity estimate for *P. cyclostomus* and the size of maturity for *P. barberinus* by Prince (2016b) was based on samples from only two fish. Given the popularity of these species in the market, it is recommended that size of maturity is estimated based on gonad examination so that these stocks can be more accurately evaluated in the future.

Haemulidae (sweetlips)

No minimum size limits were proposed for species from the Haemulidae family in the KSLCFMP. *P. lineatus* was assessed as depleting, and had an SPR less than 0.3, with a relatively high percentage of immature fish landed. Therefore a minimum size limit of 414 mm is suggested to avoid the SPR decreasing further. *P. albovittatus* was assessed as sustainable, however this species is listed as being very vulnerable with a very low resilience on FishBase due to its large size. Life history parameters were not available for this species from Palau, with only four LBSPR scenarios run. It is

recommended that size of maturity is estimated based on gonad examination so that this species can be more accurately assessed in the future.

Scombridae and Holocentridae (mackerels and squirrelfish)

No minimum size limits were proposed for species from the Scombridae or Holocentridae family in the KSLCFMP. *R. kanagurta* and *S. spiniferum* were assessed as being depleting, however there is no locally derived size of maturity information for these species. *R. kanagurta* was uncommon in the 2023 survey, however *S. spiniferum* was a popular species and could benefit from the estimation of life history parameters from Palau.

4.3 Limitations of the survey

Although LB-SPR can be a useful tool to indicate how fishing is affecting the spawning potential of fish populations, there are several limitations to this method. One factor that can affect the LB-SPR results is low sample size. Ideally, the number of length measurements for each species and year should be >1000 (Hordyk et al., 2014). However, in data-poor fisheries, such as Palau's, this is not normally possible, and many species and years had much lower sample sizes which could have affected the accuracy of the LB-SPR results. In addition, the LB-SPR assessment assumes that the lengths accurately represent the adult part of a stock and that the size structure is normally distributed, however some species had immature individuals and a skewed distribution.

Life history information, including maturity parameters (L_{50} and L_{95}) and growth parameters (M/k and L_{inf}) are needed as inputs in the LB-SPR assessment, however these parameters can vary substantially between geographical regions, with water temperature and latitude (Taylor et al., 2021). It is therefore important for this information to be collected from the same geographical region where the fish are being assessed, since LB-SPR is highly sensitive to these inputs, especially L_{inf} and M/k . Life history parameters were not available from Palau for numerous species. Size of maturity was previously estimated in the northern reefs; however this was based on macroscopic staging which is not as accurate as staging using histological techniques (Vitale et al., 2006). Furthermore, size of maturity for several important species, including *N. unicornis*, *A. nigricauda*, and *P. barberinus*, was based on a very low

sample size (14, 5, and 2, respectively), and a size of maturity curve was not able to be created (Prince, 2016b). More recently, L_{50} and L_{95} were estimated for some key species using histology (Prince et al., 2022), however there are numerous species that are important in Palau's reef fishery which still have no size of maturity information (see Appendix 3). In addition, L_{inf} and M/k inputs are only available from Palau for four species (Moore et al., 2014), with the rest estimated using empirical equations or data from numerous species and regions, which may affect the accuracy of the LB-SPR results. L_{inf} was estimated using our landing data (Powell-Wetherall plots and Froese equations), however if these species had already been overfished then L_{inf} will have been underestimated, leading to higher SPR values. This highlights the need for additional life history parameters to be estimated for key reef fish species in Palau, which will allow us to have more confidence in the LB-SPR results. In response to this, PICRC has recently partnered with The Nature Conservancy (TNC) to start collecting gonads, otoliths, and length data for some key reef fish species to obtain this information. The Coral Reef Research Foundation (CRRF) also has otoliths which were previously collected in Palau and these samples will also be used in the analysis.

Some other limitations for the fish landing surveys include the rejection of small fish before videos were taken, as well as the market rejecting fish if they already have enough of a certain species. In 2023, fish were first selected at the dock and then brought inside the JR5 market where they were filmed. This meant that small fish which were usually rejected at the dock, were not included in the survey, which will have affected the percentage of fish below the size of maturity. In 2017, fish were videoed as they were offloaded at the dock, however because the fish were thrown directly into baskets, this meant that many fish were overlapping or distorted and could not be measured. It is not known whether small fish were rejected before or after measurements were made in 2015. In addition, some fishers had already sold part of their catch before selling their fish to the market in 2017 and 2023, and therefore the total catch value used in the CPUE estimates was not always correct, which may have affected the CPUE results.

4.4 Recommendations and conclusions

Despite the limitations described above, there are several recommendations and conclusions that can be made based on the data analysed. It is recommended that the

minimum size limits for the five grouper species are increased based on the most updated size of maturity information, since the current size limit of 12 inches is not protecting most of the immature fish for most species. We also recommend updating the proposed minimum size limits in the KSLCFMP based on the presented data and most recent size of maturity information, as well as adding minimum size limits for some key reef fish species which were not included such as *A. nigricauda*, *L. xanthochilus*, *M. grandoculis*, and *V. louti*. Once life history parameters are estimated, based on data currently being collected by PICRC and TNC, minimum size limits can be updated and/or added for additional species if deemed necessary. Although our results showed some species were sustainable, there was a large uncertainty in the estimation of SPR and F/M for most species and therefore a precautionary approach should be taken. It is recommended that current management initiatives, such as the reef fish export ban, are improved (by improving enforcement) and additional management measures, such as a more conservative minimum size limit and protection of additional aggregation sites, are implemented.

It is recommended that another year-long survey is conducted at the JR5 fish market in a few years' time (e.g. 3-5 years) using similar methods as the 2023 survey, and a long-term monitoring program is set up to continue monitoring reef fish stocks over time. In addition, alternative monitoring approaches, such as intercepting fishers as they return to other landing sites as done by the Guam Division of Aquatic and Wildlife Resources (Houk et al., 2018), could be introduced to try and monitor the subsistence fishery, which is largely unaccounted for in Palau. Furthermore, community outreach and education initiatives should be implemented to increase support for fisheries management measures and reduce the occurrence of illegal fishing activities, as well as efforts made to improve enforcement.

Acknowledgements

We would like to thank current and previous PICRC staff members and interns who assisted with data collection at the JR5 fish market in 2016-2017 and 2023-2024. We also thank the owner and staff of the JR5 fish market for allowing us to collect this vital data. We are grateful to BOF and SPC for giving us access to their data from 2014-2015. The 2023-2024 survey was made possible with support from the Great Barrier

Reef Foundation's Resilient Reefs Initiative. The 2016-2017 survey was funded by the University of Guam. Fish images were created by Hiromi Ito from PICRC.

References

- BEETS, J. & FRIEDLANDER, A. 1999. Evaluation of a conservation strategy: A spawning aggregation closure for red hind, *Epinephelus guttatus*, in the U.S. Virgin Islands. *Environmental Biology of Fishes*, 55, 91-98.
- BEJARANO, S., GOLBUU, Y., SAPOLU, T. & MUMBY, P. 2013a. Ecological risk and the exploitation of herbivorous reef fish across Micronesia. *Marine Ecology Progress Series*, 482, 197-215.
- BEJARANO, S., MUMBY, P. & GOLBUU, Y. 2013b. Changes in the spear fishery of herbivores associated with closed grouper season in Palau, Micronesia. *Animal Conservation*, 17.
- BOETTIGER, C., TEMPLE LANG, D. & WAINWRIGHT, P. 2012. Rfishbase: Exploring, manipulating and visualizing FishBase data from R. *Journal of fish biology*, 81, 2030-9.
- BONALDO, R., HOEY, A. & BELLWOOD, D. 2014. The Ecosystem Roles of Parrotfishes on Tropical Reefs. *Oceanography and marine biology*, 52, 81-132.
- COSTELLO, C., OVANDO, D., HILBORN, R., GAINES, S., DESCHENES, O. & LESTER, S. 2012. Status and Solutions for the World's Unassessed Fisheries. *Science*, 338, 517-520.
- DACKS, R., LEWIS, S. A., JAMES, P. A. S., MARINO, L. L. & OLESON, K. L. L. 2020. Documenting baseline value chains of Palau's nearshore and offshore fisheries prior to implementing a large-scale marine protected area. *Marine Policy*, 117.
- ETPISON, M. T. & COLIN, P. L. 2018. Blue Water Spawning by Moorish Idols and Orangespine Surgeonfish in Palau: Is it a "Suicide Mission"? *aqua*, 23, 121.
- FILIOUS, A., DAXBOECK, C., BEGUET, T. & COOK, C. 2022. The life history of longnose emperors (*Lethrinus olivaceus*) and a data-limited assessment of their stock to support fisheries management at Rangiroa Atoll, French Polynesia. *Journal of Fish Biology*, 100, 632-644.
- FROESE, R. 2004. Keep it simple: Three indicators to deal with overfishing. *Fish and Fisheries*, 5, 86-91.
- FROESE, R. & BINOHLAN, C. 2000. Empirical relationships to estimate asymptotic length, length at first maturity and length at maximum yield per recruit in fishes, with a simple method to evaluate length frequency data. *Journal of Fish Biology*, 56, 758-773.
- FROESE, R. & PAULY, D. 2024. *FishBase* [Online]. Available: www.fishbase.org [Accessed].
- FROESE, R., WINKER, H., CORO, G., DEMIREL, N., TSIKLIRAS, A. C., DIMARCHOPOULOU, D., SCARCELLA, G., PROBST, W. N., DUREUIL, M. & PAULY, D. 2018. A new approach for estimating stock status from length frequency data. *ICES Journal of Marine Science*, 76, 350-351.
- GILLETT, R. & MOY, W. 2006. Spearfishing in the Pacific Islands: current status and management issues. Rome: FAO.
- GOODYEAR, C. P. 1993. Spawning Stock Biomass per Recruit in Fisheries Management: Foundation and Current Use.

- GOUEZO, M., GIBBONS-DECHERONG, L. & AZUMA-MALSOL, M. 2021. Koror Southern Lagoon Coastal Fisheries Management Plan 2021.
- GRAHAM, T. 2001. The Live Reef Fisheries of Palau: History and Prospects for Management. *Asia Pacific Coastal Marine Program Report #0103*. Honolulu, Hawaii, USA: The Nature Conservancy.
- GREEN, A. L. & BELLWOOD, D. R. 2009. Monitoring functional groups of herbivorous reef fishes as indicators of coral reef resilience. *IUCN Resilience Science Group Working Paper Series – No 7*. IUCN.
- GRUSS, A., ROBINSON, J., HEPPELL, S. S., HEPPELL, S. A. & SEMMENS, B. X. 2014. Conservation and fisheries effects of spawning aggregation marine protected areas: What we know, where we should go, and what we need to get there. *Ices Journal of Marine Science*, 71, 1515-1534.
- HALPERN, B. S. & WARNER, R. R. 2002. Marine reserves have rapid and lasting effects. *Ecology Letters*, 5, 361-366.
- HARDING, S., MARAMA, K., BRECKWOLDT, A., MATAIRAKULA, U. & FACHE, E. 2022. Marine resources and their value in Kadavu, Fiji. *Ambio*, 51, 2414-2430.
- HEPPELL, S. A., SEMMENS, B. X., ARCHER, S. K., PATTENGILL-SEMMENS, C. V., BUSH, P. G., MCCOY, C. M., HEPPELL, S. S. & JOHNSON, B. C. 2012. Documenting recovery of a spawning aggregation through size frequency analysis from underwater laser calipers measurements. *Biological Conservation*, 155, 119-127.
- HORDYK, A. 2021. *LBSPR: Length-Based Spawning Potential Ratio. R package version 0.1.6* [Online]. Available: <https://CRAN.R-project.org/package=LBSPR> [Accessed].
- HORDYK, A., ONO, K., VALENCIA, S., LONERAGAN, N. & PRINCE, J. 2014. A novel length-based empirical estimation method of spawning potential ratio (SPR), and tests of its performance, for small-scale, data-poor fisheries. *ICES Journal of Marine Science*.
- HORDYK, A. R., LONERAGAN, N. R. & PRINCE, J. D. 2015. An evaluation of an iterative harvest strategy for data-poor fisheries using the length-based spawning potential ratio assessment methodology. *Fisheries Research*, 171, 20-32.
- HOUK, P., CUETOS-BUENO, J., TIBBATTIS, B. & GUTIERREZ, J. 2018. Variable density dependence and the restructuring of coral-reef fisheries across 25 years of exploitation. *Scientific Reports*, 8.
- HOUK, P., RHODES, K., CUETOS-BUENO, J., LINDFIELD, S., FREAD, V. & MCILWAIN, J. 2011. Commercial coral-reef fisheries across Micronesia: A need for improving management. *Coral Reefs*, 31, 13-26.
- HSU, T.-H., ADIPUTRA, Y. T., BURRIDGE, C. P. & GWO, J.-C. 2011. Two spinefoot colour morphs: mottled spinefoot *Siganus fuscescens* and white-spotted spinefoot *Siganus canaliculatus* are synonyms. *Journal of Fish Biology*, 79, 1350-1355.
- JENSEN, A. L. 1996. Beverton and Holt life history invariants result from optimal trade-off of reproduction and survival. *Canadian Journal of Fisheries and Aquatic Sciences*, 53, 820-822.
- JOHANNES, R. E. 1981. *Words of the Lagoon: Fishing and Marine Lore in the Palau District of Micronesia*, University of California Press.
- JOHANNES, R. E. 1991. Some suggested management initiatives in Palau's nearshore fisheries, and the relevance of traditional management. Noumea, New Caledonia: South Pacific Commission.

- JOHANNES, R. E., SQUIRE, L., GRAHAM, T., SADOVY, Y. & RENGUUL, H. 1999. Spawning aggregations of groupers (Serranidae) in Palau. *The Nature Conservancy Marine Research Series Publication No. 1*. The Nature Conservancy.
- KITALONG, A. & DALZELL, P. 1994. A preliminary assessment of the status of inshore coral reef fish stocks in Palau. Noumea, New Calidonia: South Pacific Commission.
- KSG 2012. Rock Islands Southern Lagoon Management Plan 2012-2016. Koror State Government Republic of Palau.
- LINDFIELD, S. 2020. Validating the use of underwater stereo-video for determining life-history parameters of aggregation spawning coral reef fishes. *Final report: Saltonstall-Kennedy Competitive Research Program*. Coral Reef Research Foundation.
- LINDFIELD, S. J. 2017. Palau's reef fisheries: changes in size and spawning potential from past to present. Coral Reef Research Foundation.
- LINGARD, S., HARPER, S., OTA, Y. & ZELLER, D. 2011. Marine Fisheries of Palau, 1950-2008: Total Reconstructed Catch. In: HARPER, S. & ZELLER, D. (eds.) *Fisheries catch reconstructions: Islands, Part II*. Fisheries Centre, University of British Columbia.
- MACNEIL, M. A., GRAHAM, N. A. J., CINNER, J. E., WILSON, S. K., WILLIAMS, I. D., MAINA, J., NEWMAN, S., FRIEDLANDER, A. M., JUPITER, S., POLUNIN, N. V. C. & MCCLANAHAN, T. R. 2015. Recovery potential of the world's coral reef fishes. *Nature*, 520, 341-344.
- MAUNDER, M. N., SIBERT, J. R., FONTENEAU, A., HAMPTON, J., KLEIBER, P. & HARLEY, S. J. 2006. Interpreting catch per unit effort data to assess the status of individual stocks and communities. *ICES Journal of Marine Science*, 63, 1373-1385.
- MEDEIROS-LEAL, W., SANTOS, R., PEIXOTO, U. I., CASAL-RIBEIRO, M., NOVOA-PABON, A., SIGLER, M. F. & PINHO, M. 2023. Performance of length-based assessment in predicting small-scale multispecies fishery sustainability. *Reviews in Fish Biology and Fisheries*, 33, 819-852.
- MILDENBERGER, T. & TAYLOR, M. 2016. TropFishR: an R package for fisheries analysis with length-frequency data.
- MOORE, B., RECHELLUUL, P. & VICTOR, S. 2014. Creel survey and demographic assessments of coastal finfish fisheries of southern Palau. Secretariat of the Pacific Community.
- MOORE, B. R. 2019. Age-based life history of humpback red snapper, *Lutjanus gibbus*, in New Caledonia. *Journal of Fish Biology*, 95, 1374-1384.
- MULLER-KARANASSOS, C., FILOUS, A., FRIEDLANDER, A. M., CUETOS-BUENO, J., GOUZO, M., LINDFIELD, S. J., NESTOR, V., MARINO, L. L., MEREB, G., OLSUDONG, D. & GOLBUU, Y. 2021. Effects of habitat, fishing, and fisheries management on reef fish populations in Palau. *Fisheries Research*, 241.
- MULLER-KARANASSOS, C., OTTO, E. I., NESTOR, V., TELLEI, E., CLAASSENS, L., OLSUDONG, D., MESENGEI, M., MEREB, G. & BUKURROU, A. 2023. Reef fish stock assessment in 2021 indicates recovery over time. Palau International Coral Reef Center.
- NADON, M., AULT, J., WILLIAMS, I., SMITH, S. & DINARDO, G. 2015. Length-Based Assessment of Coral Reef Fish Populations in the Main and Northwestern Hawaiian Islands. *PLoS ONE*.

- NANNINGA, G. B. & SPAET, J. L. Y. 2017. Spawning aggregations of the Humpback red snapper, *Lutjanus gibbus*, in the Tuamotus, French Polynesia. *Marine Biodiversity*, 47, 375-376.
- PERRON, F. E., NARUO, A. M. & PATRIS, S. J. 1983. The Palau reef fish production study: A baseline study of the commercial reef fishing industry in Palau, and a blueprint for the development of a permanent fisheries management system. Division of Marine Resources, Republic of Palau.
- PRINCE, J. 2016a. Estimation of SPR based minimum lengths for Indo-Pacific coral reef fish in Palau. The Nature Conservancy.
- PRINCE, J. 2016b. Length based SPR assessment of eighteen Indo-Pacific coral reef fish populations in Palau. The Nature Conservancy.
- PRINCE, J., HARFORD, W. J., TAYLOR, B. M. & LINDFIELD, S. J. 2022. Standard histological techniques systematically under-estimate the size fish start spawning. *Fish and Fisheries*, 23, 1507-1516.
- PRINCE, J. & HORDYK, A. 2018. What to do when you have almost nothing: A simple quantitative prescription for managing extremely data-poor fisheries. *Fish and Fisheries*, 20.
- PRINCE, J., HORDYK, A., VALENCIA, S., LONERAGAN, N. & SAINSBURY, K. 2014. Revisiting the concept of Beverton-Holt Life History Invariants with the aim of informing data-poor fisheries assessment. *ICES Journal of Marine Science*.
- PRINCE, J. & LINDFIELD, S. 2018. Estimating size at maturity using length measurements from spawning aggregations and deriving 20% SPR size limits for at least 5 reef fish species. Biospherics and Coral Reef Research Foundation.
- PRINCE, J., VICTOR, S., KLOULCHAD, V. & HORDYK, A. 2015. Length based SPR assessment of eleven Indo-Pacific coral reef fish populations in Palau. *Fisheries Research*, 171, 42-58.
- PRINCE, J. D., WILCOX, C. & HALL, N. 2023. How to estimate life history ratios to simplify data-poor fisheries assessment. *ICES Journal of Marine Science*, 80, 2619-2629.
- RUSS, G. R. & ALCALA, A. C. 2004. Marine reserves: long-term protection is required for full recovery of predatory fish populations. *Oecologia*, 138, 622-7.
- SADOVY DE MITCHESON, Y. & COLIN, P. L. 2012. *Reef Fish Spawning Aggregations: Biology, Research and Management*, Springer Netherlands.
- SADOVY DE MITCHESON, Y., COLIN, P. L., LINDFIELD, S. J. & BUKURROU, A. 2020. A decade of monitoring an Indo-Pacific grouper spawning aggregation: benefits of protection and importance of survey design. *Frontiers in Marine Science*, 7, 571878.
- SADOVY, Y. 2005. Trouble on the reef: the imperative for managing vulnerable and valuable fisheries. *Fish and Fisheries*, 6, 167-185.
- SADOVY, Y. 2007. Report on current status and exploitation history of reef fish spawning aggregations in Palau. *Western Pacific Fishery Survey Series: Society for the Conservation of Reef Fish Aggregations, Volume 3*. Society for the Conservation of Reef Fish Aggregations and Palau Conservation Society.
- SADOVY, Y. & DOMEIER, M. 2005. Are aggregation-fisheries sustainable? Reef fish fisheries as a case study. *Coral Reefs*, 24, 254-262.
- SAKAUE, J., AKINO, H., ENDO, M., IDA, H. & ASAHIDA, T. 2016. Temporal and spatial site sharing during spawning in snappers *Symphoricthys spilurus* and

- Lutjanus bohar (Pisces: Perciformes: Lutjanidae) in waters around Peleliu Island, Palau. *Zoological Studies*, 55, 15.
- SHANTZ, A. A., LADD, M. C. & BURKEPILE, D. E. 2020. Overfishing and the ecological impacts of extirpating large parrotfish from Caribbean coral reefs. *Ecological Monographs*, 90, e01403.
- TAYLOR, B., WAKEFIELD, C., NEWMAN, S., CHINKIN, M. & MEEKAN, M. 2021. Unprecedented longevity of unharvested shallow-water snappers in the Indian Ocean. *Coral Reefs*, 40.
- TAYLOR, B. M., GOURLEY, J. & TRIANNI, M. S. 2017. Age, growth, reproductive biology and spawning periodicity of the forktail rabbitfish (<i>Siganus argenteus</i>) from the Mariana Islands. *Marine and Freshwater Research*, 68, 1088-1097.
- TNC 2016. Palau's Northern Reef Fisheries Management Plan 2016. The Nature Conservancy.
- VASILAKOPOULOS, P., O'NEILL, F. G. & MARSHALL, C. T. 2011. Misspent youth: does catching immature fish affect fisheries sustainability? *ICES Journal of Marine Science*, 68, 1525-1534.
- VITALE, F., SVEDÄNG, H. & CARDINALE, M. 2006. Histological analysis invalidates macroscopically determined maturity ogives of the Kattegat cod (*Gadus morhua*) and suggests new proxies for estimating maturity status of individual fish. *ICES Journal of Marine Science*, 63, 485-492.

Appendices

Appendix 1 – Current, proposed and recommended minimum size limits (fork length) for reef fish species in Palau. Sizes are in mm. Species that were not assessed do not have size of maturity information from Palau.

Species	Palau-wide	Ngarchelong	Kayang el	Koror (proposed)	Recommended size limit (L ₅₀ x 1.1)	Recommended size limit (L ₅₀ x 1.2)
<i>Acanthurus nigricauda</i>	None	None	None	None	209	228
<i>Acanthurus xanthopterus</i>	None	None	None	254	Not assessed	Not assessed
<i>Cetoscarus bicolor/Cetoscarus ocellatus</i>	None	279	330	356	351	383
<i>Cetoscarus bleekeri</i>	None	None	None	305	Not assessed	Not assessed
<i>Chlorurus japanensis</i>	None	None	None	203	Not assessed	Not assessed
<i>Chlorurus microrhinus/Chlorurus frontalis</i>	None	279	330	356	340	371
<i>Epinephelus polyphekadion</i>	305 (during open season)	356 (during open season)	406 (during open season)	None	393	428
<i>Epinephelus fuscoguttatus</i>	305 (during open season)	356 (during open season)	406 (during open season)	None	619	676

<i>Hipposcarus harid</i>	None	None	None	356	None - this species is not present in Palau	None - this species is not present in Palau
<i>Hipposcarus longiceps</i>	None	254	305	305	276	301
<i>Lutjanus bohar</i>	None	406	457	457 (during open season)	442	482
<i>Lutjanus gibbus</i>	None	254	305	254 (after 6-month moratorium)	270	294
<i>Lethrinus obsoletus</i>	None	None	None	None	233	254
<i>Lethrinus olivaceus</i>	None	406	457	457	465	508
<i>Lethrinus xanthochilus</i>	None	None	330	None	337	367
<i>Monotaxis grandoculis</i>	None	None	None	None	341	372
<i>Naso unicornis</i>	None	None	406	356	399	436
<i>Naso lituratus</i>	None	None	None	203	179	196
<i>Plectorhinchus lineatus</i>	None	None	None	None	414	451
<i>Plectropomus areolatus</i>	305 (during open season)	356 (during open season)	406 (during open season)	None	420	458
<i>Plectropomus leopardus</i>	305 (during open season)	279 (during open season)	330 (during open season)	None	320	349
<i>Plectropomus laevis</i>	305 (during open season)	559 (during open season)	610 (during open season)	None	Not assessed	Not assessed
<i>Scarus oviceps</i>	None	None	None	203	Not assessed	Not assessed
<i>Scarus altipinnis</i>	None	None	None	305	Not assessed	Not assessed
<i>Scarus chameleon</i>	None	None	None	305	Not assessed	Not assessed
<i>Scarus prasiognathos</i>	None	None	None	305	Not assessed	Not assessed
<i>Scarus psittacus</i>	None	None	None	203	Not assessed	Not assessed
<i>Scarus flavipectoralis</i>	None	None	None	203	Not assessed	Not assessed
<i>Scarus rubroviolaceus</i>	None	None	None	305	321	350
<i>Scarus schlegeli</i>	None	None	None	203	Not assessed	Not assessed
<i>Scarus spinus</i>	None	None	None	203	Not assessed	Not assessed
<i>Scarus frenatus</i>	None	None	None	203	Not assessed	Not assessed
<i>Variola louti</i>	None	254	330	None	314	342

Appendix 2. Conversion of fork length (FL) to total length (TL) for 2009 herbivore data

Species	Formula with conversion values	Source
<i>Acanthurus nigricauda</i>	$FL=TL/1.224$	Fish Base – mean of two values
<i>Cetoscarus ocellatus</i>	$FL=TL/1.083$	Fish Base
<i>Chlorurus microrhinos</i>	$FL=TL/1.188$	Fish Base
<i>Hipposcarus longiceps</i>	$FL=TL/1.012$	Fish Base
<i>Scarus rubroviolaceus</i>	$FL=TL/1.099$	Fish Base
<i>Siganus argenteus</i>	$FL=TL/1.121$	Fish Base
<i>Siganus doliatus</i>	$FL=TL*0.947$	Fish market data 2023-2024
<i>Siganus fuscescens/canaliculatus</i>	$FL=TL*0.94$	Fish Base
<i>Siganus lineatus</i>	$FL=TL/1.041$	Fish Base
<i>Siganus punctatus</i>	$FL=TL*0.907$	Fish Base – <i>Siganus punctatissimus</i>

Appendix 3. Table of life history parameters used in the LB-SPR assessment for 45 species

Species	Perce tile_99t h	L50_ froes e	L95_ froes e	Linf_ froes e	Linf_ fi shbas e	Linf_ princ e	Linf_ powe ll	MK_pr ince_s p	MK_pri nce_fa m	L50_ princ e	L95_ princ e	MK_fi shbas e	L50_fi shbas e	L95_fishb ase_froes e	MK_ 1. 5	Linf_ moor e	MK_ moor e
<i>Acanthurus nigricauda</i>	276	138	152	263	243	238	236	0.35	0.518	190	200	NA	184	202	1.5	NA	NA
<i>Acanthurus xanthopterus</i>	469	221	243	445	426	NA	457	NA	0.518	NA	NA	NA	NA	NA	1.5	NA	NA
<i>Aethaloperca rogaa</i>	389	187	206	370	NA	NA	411	NA	1.165	NA	NA	NA	NA	NA	1.5	NA	NA
<i>Atule mate</i>	300	149	164	286	290	NA	299	NA	1.277	NA	NA	2.18	172	189	1.5	NA	NA
<i>Caranx melampygus</i>	616	281	309	581	857	NA	579	NA	1.277	286	353	NA	350	385	1.5	NA	NA
<i>Cephalopholis argus</i>	381	184	202	362	387	NA	352	NA	1.165	NA	NA	0.41	NA	NA	1.5	351	0.835
<i>Cetoscarus ocellatus</i>	445	211	232	422	421	515	506	0.7	0.94	319	345	NA	300	330	1.5	NA	NA
<i>Chlorurus microrhinos</i>	454	214	236	430	451	461	531	0.7	0.94	309	345	NA	NA	NA	1.5	NA	NA
<i>Epinephelus polyphekadion</i>	555	256	282	525	556	NA	620	NA	1.165	357	397	0.62	NA	NA	1.5	NA	NA
<i>Ferdauia orthogrammus</i>	472	222	244	448	NA	NA	524	NA	1.277	NA	NA	NA	NA	NA	1.5	NA	NA
<i>Hipposcarus longiceps</i>	411	196	216	390	409	423	495	1.07	0.94	251	273	0.64	NA	NA	1.5	NA	NA
<i>Lethrinus erythracanthus</i>	653	295	325	615	418	NA	687	NA	0.866	NA	NA	NA	NA	NA	1.5	NA	NA
<i>Lethrinus lentjan</i>	402	192	212	381	408	NA	412	NA	0.866	NA	NA	0.68	245	269	1.5	NA	NA
<i>Lethrinus obsoletus</i>	324	159	175	309	303	337	332	0.62	0.866	212	237	NA	237	261	1.5	262	0.293
<i>Lethrinus olivaceus</i>	629	286	314	593	802	579	650	0.62	0.866	423	483	0.98	370	407	1.5	NA	NA
<i>Lethrinus xanthochilus</i>	514	239	263	486	588	466	568	0.62	0.866	306	350	NA	NA	NA	1.5	NA	NA
<i>Lutjanus bohar</i>	643	291	321	606	745	579	691	0.41	0.977	402	475	1.45	477	525	1.5	NA	NA

<i>Lutjanus gibbus</i>	377	182	200	359	386	327	429	0.41	0.977	245	320	NA	212	233	1.5	285	0.4
<i>Lutjanus monostigma</i>	406	194	214	386	550	NA	439	NA	0.977	NA	NA	NA	NA	NA	1.5	NA	NA
<i>Lutjanus vitta</i>	318	156	172	303	309	NA	310	NA	0.977	NA	NA	0.77	154	169	1.5	NA	NA
<i>Monotaxis grandoculis</i>	464	219	241	440	NA	NA	503	NA	0.866	310	NA	NA	235	259	1.5	NA	NA
<i>Naso lituratus</i>	309	153	168	295	245	256	321	0.35	0.518	163	215	NA	NA	NA	1.5	242	0.151
<i>Naso unicornis</i>	531	246	271	503	480	454	509	0.35	0.518	363	420	NA	NA	NA	1.5	449	0.239
<i>Parupeneus barberinus</i>	370	179	197	352	336	354	345	1.5	1.871	230	250	NA	192	211	1.5	NA	NA
<i>Parupeneus cyclostomus</i>	360	175	192	343	NA	NA	322	NA	1.871	NA	NA	NA	NA	NA	1.5	NA	NA
<i>Plectorhinchus albobittatus</i>	773	343	377	727	NA	NA	791	NA	1.244	NA	NA	NA	NA	NA	1.5	NA	NA
<i>Plectorhinchus lineatus</i>	531	246	271	502	630	NA	597	NA	1.244	376	NA	NA	NA	NA	1.5	NA	NA
<i>Plectropomus areolatus</i>	565	260	286	534	598	619	559	0.91	1.165	382	449	1.27	368	404	1.5	NA	NA
<i>Plectropomus leopardus</i>	428	203	224	406	620	493	459	0.91	1.165	291	315	2.30	366	403	1.5	NA	NA
<i>Rastrelliger kanagurta</i>	337	165	181	321	296	NA	315	NA	0.987	NA	NA	1.71	198	217	1.5	NA	NA
<i>Sargocentron spiniferum</i>	322	158	174	307	533	NA	363	NA	NA	NA	NA	2.04	NA	NA	1.5	NA	NA
<i>Scarus forsteni</i>	425	202	222	403	281	NA	362	NA	0.94	NA	NA	NA	NA	NA	1.5	NA	NA
<i>Scarus frenatus</i>	350	170	187	333	231	NA	342	NA	0.94	NA	NA	0.24	NA	NA	1.5	NA	NA
<i>Scarus ghobban</i>	485	227	250	459	286	NA	551	NA	0.94	NA	NA	0.58	NA	NA	1.5	NA	NA
<i>Scarus prasiognathos</i>	404	193	213	384	NA	NA	398	NA	0.94	NA	NA	NA	NA	NA	1.5	NA	NA
<i>Scarus rubroviolaceus</i>	436	207	228	414	387	406	456	0.36	0.94	292	390	0.73	NA	NA	1.5	NA	NA

<i>Scarus schlegeli</i>	328	161	177	313	283	NA	315	NA	0.94	NA	NA	1.21	NA	NA	1.5	NA	NA
<i>Siganus argenteus</i>	298	148	163	285	310	NA	319	NA	1.651	NA	NA	NA	NA	NA	1.5	NA	NA
<i>Siganus corallinus</i>	230	117	129	220	NA	NA	232	NA	1.651	NA	NA	NA	NA	NA	1.5	NA	NA
<i>Siganus doliatus</i>	248	126	138	238	NA	NA	236	NA	1.651	NA	NA	NA	NA	NA	1.5	NA	NA
<i>Siganus fuscescens</i>	234	119	131	224	250	NA	245	NA	1.651	NA	NA	NA	53	58	1.5	NA	NA
<i>Siganus lineatus</i>	328	161	177	313	336	375	322	1.5	1.651	244	250	NA	NA	NA	1.5	NA	NA
<i>Siganus puellus</i>	258	130	143	247	NA	NA	263	NA	1.651	NA	NA	NA	NA	NA	1.5	NA	NA
<i>Siganus punctatus</i>	305	151	166	291	NA	NA	316	NA	1.651	222	230	NA	NA	NA	1.5	NA	NA
<i>Variola louti</i>	446	211	232	423	NA	483	492	0.86	1.165	285	350	1.54	NA	NA	1.5	NA	NA

Appendix 4. LB-SPR scenarios run for 45 species

Scenario	Linf_name	MK_name	L50_name	L95_name
1	Linf_froese	MK_1.5	L50_froese	L95_froese
2	Linf_prince	MK_1.5	L50_froese	L95_froese
3	Linf_fishbase	MK_1.5	L50_froese	L95_froese
4	Linf_froese	MK_prince_fam	L50_froese	L95_froese
5	Linf_prince	MK_prince_fam	L50_froese	L95_froese
6	Linf_fishbase	MK_prince_fam	L50_froese	L95_froese
7	Linf_froese	MK_fishbase	L50_froese	L95_froese
8	Linf_fishbase	MK_fishbase	L50_froese	L95_froese
9	Linf_froese	MK_1.5	L50_fishbase	L95_fishbase_froese
10	Linf_prince	MK_1.5	L50_fishbase	L95_fishbase_froese
11	Linf_fishbase	MK_1.5	L50_fishbase	L95_fishbase_froese
12	Linf_froese	MK_prince_fam	L50_fishbase	L95_fishbase_froese
13	Linf_prince	MK_prince_fam	L50_fishbase	L95_fishbase_froese
14	Linf_fishbase	MK_prince_fam	L50_fishbase	L95_fishbase_froese
15	Linf_froese	MK_fishbase	L50_fishbase	L95_fishbase_froese
16	Linf_fishbase	MK_fishbase	L50_fishbase	L95_fishbase_froese
17	Linf_froese	MK_1.5	L50_prince	L95_prince
18	Linf_prince	MK_1.5	L50_prince	L95_prince
19	Linf_fishbase	MK_1.5	L50_prince	L95_prince
20	Linf_froese	MK_prince_fam	L50_prince	L95_prince
21	Linf_prince	MK_prince_fam	L50_prince	L95_prince
22	Linf_fishbase	MK_prince_fam	L50_prince	L95_prince
23	Linf_froese	MK_fishbase	L50_prince	L95_prince
24	Linf_fishbase	MK_fishbase	L50_prince	L95_prince
25	Linf_froese	MK_prince_sp	L50_prince	L95_prince
26	Linf_prince	MK_prince_sp	L50_prince	L95_prince
27	Linf_fishbase	MK_prince_sp	L50_prince	L95_prince
28	Linf_froese	MK_prince_sp	L50_froese	L95_froese
29	Linf_prince	MK_prince_sp	L50_froese	L95_froese
30	Linf_fishbase	MK_prince_sp	L50_froese	L95_froese
31	Linf_froese	MK_prince_sp	L50_fishbase	L95_fishbase_froese
32	Linf_prince	MK_prince_sp	L50_fishbase	L95_fishbase_froese
33	Linf_fishbase	MK_prince_sp	L50_fishbase	L95_fishbase_froese
34	Linf_powell	MK_1.5	L50_froese	L95_froese

35	Linf_powell	MK_prince_fa m	L50_froese	L95_froese
36	Linf_powell	MK_prince_sp	L50_froese	L95_froese
37	Linf_powell	MK_1.5	L50_fishbas e	L95_fishbase_froese
38	Linf_powell	MK_prince_fa m	L50_fishbas e	L95_fishbase_froese
39	Linf_powell	MK_prince_sp	L50_fishbas e	L95_fishbase_froese
40	Linf_powell	MK_1.5	L50_prince	L95_prince
41	Linf_powell	MK_prince_fa m	L50_prince	L95_prince
42	Linf_powell	MK_prince_sp	L50_prince	L95_prince
43	Linf_moore	MK_moore	L50_froese	L95_froese
44	Linf_moore	MK_moore	L50_fishbas e	L95_fishbase_froese
45	Linf_moore	MK_moore	L50_prince	L95_prince
46	Linf_moore	MK_1.5	L50_froese	L95_froese
47	Linf_moore	MK_1.5	L50_fishbas e	L95_fishbase_froese
48	Linf_moore	MK_1.5	L50_prince	L95_prince
49	Linf_moore	MK_prince_fa m	L50_froese	L95_froese
50	Linf_moore	MK_prince_fa m	L50_fishbas e	L95_fishbase_froese
51	Linf_moore	MK_prince_fa m	L50_prince	L95_prince
52	Linf_moore	MK_prince_sp	L50_froese	L95_froese
53	Linf_moore	MK_prince_sp	L50_fishbas e	L95_fishbase_froese
54	Linf_moore	MK_prince_sp	L50_prince	L95_prince
55	Linf_prince	MK_fishbase	L50_froese	L95_froese
56	Linf_froese	MK_moore	L50_froese	L95_froese
57	Linf_prince	MK_fishbase	L50_fishbas e	L95_fishbase_froese
58	Linf_froese	MK_moore	L50_fishbas e	L95_fishbase_froese
59	Linf_prince	MK_fishbase	L50_prince	L95_prince
60	Linf_froese	MK_moore	L50_prince	L95_prince
61	Linf_powell	MK_fishbase	L50_froese	L95_froese
62	Linf_prince	MK_moore	L50_froese	L95_froese
63	Linf_powell	MK_fishbase	L50_fishbas e	L95_fishbase_froese
64	Linf_prince	MK_moore	L50_fishbas e	L95_fishbase_froese
65	Linf_powell	MK_fishbase	L50_prince	L95_prince
66	Linf_prince	MK_moore	L50_prince	L95_prince
67	Linf_moore	MK_fishbase	L50_froese	L95_froese
68	Linf_fishbas e	MK_moore	L50_froese	L95_froese
69	Linf_moore	MK_fishbase	L50_fishbas e	L95_fishbase_froese
70	Linf_fishbas e	MK_moore	L50_fishbas e	L95_fishbase_froese
71	Linf_moore	MK_fishbase	L50_prince	L95_prince
72	Linf_fishbas e	MK_moore	L50_prince	L95_prince

73	Linf_powell	MK_moore	L50_froese	L95_froese
74	Linf_powell	MK_moore	L50_fishbase	L95_fishbase_froese
75	Linf_powell	MK_moore	L50_prince	L95_prince

Appendix 5. Results of the LB-SPR assessment for 45 species (best scenarios)

Species	Scenario	Linf_in	MK_in	L50_in	L95_in	FM	SP R	SPR_lo wer	SPR_up per	FM_lo wer	FM_up per	Linf_na me	MK_name	L50_na me	L95_na me
<i>Acanthurus nigricauda</i>	25	263	0.35	190	200	8.49	0.15	0.14	0.16	7.79	9.19	Linf_fro ese	MK_prince sp	L50_pri nce	L95_pri nce
<i>Acanthurus xanthopterus</i>	35	457	0.52	221	243	7.85	0.32	0.25	0.39	5.01	10.69	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Aethaloperca rogaa</i>	35	404	1.17	185	204	1.75	0.48	0.40	0.57	0.84	2.66	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Atule mate</i>	35	299	1.28	149	163	5.07	0.80	0.72	0.88	1.65	8.49	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Caranx melampygus</i>	20	581	1.28	286	353	0.60	0.56	0.39	0.72	0.18	1.02	Linf_fro ese	MK_prince fam	L50_pri nce	L95_pri nce
<i>Cephalopholis argus</i>	56	361	0.84	183	201	2.36	0.48	0.44	0.53	1.67	3.05	Linf_fro ese	MK_moore	L50_fro ese	L95_fro ese
<i>Cetoscarus ocellatus</i>	42	499	0.70	319	345	4.99	0.10	0.09	0.11	4.32	5.66	Linf_po well	MK_prince sp	L50_pri nce	L95_pri nce
<i>Chlorurus microrhinos</i>	42	527	0.70	309	345	5.72	0.06	0.05	0.07	5.01	6.43	Linf_po well	MK_prince sp	L50_pri nce	L95_pri nce
<i>Epinephelus polyphekadion</i>	41	622	1.17	357	397	3.41	0.11	0.08	0.14	2.53	4.29	Linf_po well	MK_prince fam	L50_pri nce	L95_pri nce
<i>Ferdauia orthogrammus</i>	35	476	1.28	222	244	1.53	0.57	0.43	0.70	0.36	2.70	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Hipposcarus longiceps</i>	42	496	1.07	251	273	3.91	0.16	0.15	0.16	3.74	4.08	Linf_po well	MK_prince sp	L50_pri nce	L95_pri nce
<i>Lethrinus erythracanthus</i>	35	687	0.87	295	325	1.15	0.31	0.17	0.44	0.52	1.78	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Lethrinus lentjan</i>	35	412	0.87	192	211	1.73	0.33	0.27	0.39	1.23	2.23	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Lethrinus obsoletus</i>	75	332	0.29	212	237	10.95	0.08	0.07	0.10	8.49	13.41	Linf_po well	MK_moore	L50_pri nce	L95_pri nce
<i>Lethrinus olivaceus</i>	42	650	0.62	423	483	4.24	0.13	0.10	0.16	3.29	5.19	Linf_po well	MK_prince sp	L50_pri nce	L95_pri nce
<i>Lethrinus xanthochilus</i>	42	568	0.62	306	350	5.28	0.11	0.09	0.12	4.49	6.07	Linf_po well	MK_prince sp	L50_pri nce	L95_pri nce
<i>Lutjanus bohar</i>	42	692	0.41	402	475	4.69	0.07	0.05	0.08	3.91	5.47	Linf_po well	MK_prince sp	L50_pri nce	L95_pri nce
<i>Lutjanus gibbus</i>	75	421	0.40	245	320	8.21	0.05	0.05	0.06	7.74	8.68	Linf_po well	MK_moore	L50_pri nce	L95_pri nce
<i>Lutjanus monostigma</i>	35	439	0.98	194	214	1.87	0.41	0.31	0.50	0.82	2.92	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese

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<i>Lutjanus vitta</i>	35	310	0.98	156	172	2.3 1	0.3 8	0.32	0.45	1.44	3.18	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Monotaxis grandoculis</i>	41	504	0.87	310	409	4.0 7	0.2 0	0.13	0.27	2.39	5.75	Linf_po well	MK_prince fam	L50_pri nce	L95_pri nce
<i>Naso lituratus</i>	75	320	0.15	163	215	36. 02	0.0 4	0.04	0.04	32.75	39.29	Linf_po well	MK moore	L50_pri nce	L95_pri nce
<i>Naso unicornis</i>	75	507	0.24	363	420	8.5 4	0.1 0	0.08	0.12	7.07	10.01	Linf_po well	MK moore	L50_pri nce	L95_pri nce
<i>Parupeneus barberinus</i>	28	350	1.50	178	196	0.6 6	0.7 3	0.70	0.76	0.53	0.79	Linf_fro ese	MK_prince sp	L50_fro ese	L95_fro ese
<i>Parupeneus cyclostomus</i>	4	343	1.87	175	192	0.4 7	0.8 5	0.78	0.92	0.11	0.83	Linf_fro ese	MK_prince fam	L50_fro ese	L95_fro ese
<i>Plectorhinchus albovittatus</i>	35	791	1.24	343	377	0.8 3	0.5 1	0.36	0.66	0.26	1.40	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Plectorhinchus lineatus</i>	41	548	1.24	376	402	2.3 6	0.2 7	0.21	0.33	1.57	3.15	Linf_po well	MK_prince fam	L50_pri nce	L95_pri nce
<i>Plectropomus areolatus</i>	42	559	0.91	382	449	2.3 2	0.3 0	0.20	0.40	1.24	3.40	Linf_po well	MK_prince sp	L50_pri nce	L95_pri nce
<i>Plectropomus leopardus</i>	42	459	0.91	291	315	2.7 2	0.2 1	0.19	0.24	2.29	3.15	Linf_po well	MK_prince sp	L50_pri nce	L95_pri nce
<i>Rastrelliger kanagurta</i>	4	321	0.99	165	181	2.2 8	0.6 0	0.53	0.66	1.24	3.32	Linf_fro ese	MK_prince fam	L50_fro ese	L95_fro ese
<i>Sargocentron spinfiferum</i>	34	364	1.50	157	173	3.2 3	0.4 0	0.38	0.42	2.77	3.69	Linf_po well	MK 1.5	L50_fro ese	L95_fro ese
<i>Scarus forsteni</i>	4	403	0.94	202	222	2.4 1	0.3 0	0.23	0.37	1.39	3.43	Linf_fro ese	MK_prince fam	L50_fro ese	L95_fro ese
<i>Scarus frenatus</i>	35	342	0.94	170	187	2.1 5	0.5 0	0.40	0.61	0.86	3.44	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Scarus ghobban</i>	35	551	0.94	227	250	3.0 1	0.1 8	0.16	0.21	2.41	3.61	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Scarus prasiognathos</i>	35	394	0.94	191	210	2.5 5	0.3 9	0.35	0.43	1.94	3.16	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Scarus rubroviolaceus</i>	42	454	0.36	292	390	9.5 7	0.0 8	0.07	0.08	8.59	10.55	Linf_po well	MK_prince sp	L50_pri nce	L95_pri nce
<i>Scarus schlegeli</i>	35	312	0.94	159	175	2.2 2	0.5 6	0.46	0.66	0.86	3.58	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Siganus argenteus</i>	35	306	1.65	148	162	1.0 3	0.6 2	0.59	0.65	0.85	1.21	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Siganus corallinus</i>	35	232	1.65	117	129	3.0 8	0.7 6	0.67	0.85	0.96	5.20	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Siganus doliatus</i>	4	238	1.65	126	138	0.7 6	0.8 1	0.74	0.88	0.29	1.23	Linf_fro ese	MK_prince fam	L50_fro ese	L95_fro ese

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<i>Siganus fuscescens</i>	35	245	1.65	119	131	1.55	0.70	0.69	0.72	1.31	1.79	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Siganus lineatus</i>	42	321	1.50	244	250	0.90	0.36	0.29	0.43	0.67	1.13	Linf_po well	MK_prince sp	L50_pri nce	L95_pri nce
<i>Siganus puellus</i>	35	263	1.65	130	143	1.23	0.65	0.60	0.69	0.82	1.64	Linf_po well	MK_prince fam	L50_fro ese	L95_fro ese
<i>Siganus punctatus</i>	41	314	1.65	222	230	1.02	0.41	0.38	0.44	0.89	1.15	Linf_po well	MK_prince fam	L50_pri nce	L95_pri nce
<i>Variola louti</i>	42	492	0.86	285	350	3.06	0.25	0.11	0.39	0.68	5.44	Linf_po well	MK_prince sp	L50_pri nce	L95_pri nce